Public acceptability of conservation entrance fees: Does reallocating revenues shift support before and after policy implementation?

Carlos Silva-Zambrano[1, 2, *], Claudia Aravena[1], Eoin McLaughlin[1], and Cesar Viteri[2]

[1]Heriot-Watt University, Edinburgh, United Kingdom.
[2]Charles Darwin Foundation, Galapagos Islands, Ecuador.
[*]Corresponding author: cas2002@hw.ac.uk

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Abstract

Entrance fees are widely used tools to finance conservation and manage tourism in protected areas, yet their implementation is often delayed due to public opposition. Evidence on mechanisms that increase public acceptability and on how support evolves post-policy implementation remains understudied. We address this gap by studying the public acceptability of increasing entrance fees in the Galapagos Islands, before, shortly-after and long-after policy implementation. Using a mixed-methods approach, we find that residents evaluate entrance fee policies through perceptions about transparency, fairness, expected impact on demand, and whether conservation revenues address community needs. Based on three survey waves, estimates show that despite initial and persistent opposition among residents, reallocating revenues toward publicly preferred areas substantially and increasingly improves support for the policy change shortly- and long-after implementation. We offer recommendations to enhance the political feasibility of contested policies, aimed at facilitating their timely and effective implementation in other biodiversity-rich destinations.

Keywords: Public support; Tourism taxes; Individual preferences; Nature-based tourism; Protected area governance; Biodiversity conservation

JEL Classification: H23, Q26, Q57.

1 Introduction

Protected areas are key strategies for addressing the biodiversity loss crisis, while also offering economic opportunities through nature-based tourism (Convention on Biological Diversity, 2022). However, tourism can intensify environmental pressures through negative externalities such as habitat degradation and wildlife disturbance (Monz et al., 2013). A widely used tool to mitigate these environmental pressures, especially in protected areas, is the establishment of entrance fee policies, which serve to regulate visitor flows and generate revenues that could be used for sanitation, conservation, or protected area management (Chung et al., 2011). However, the introduction and adjustment of entrance fees are frequently contested, with opposition driven by concerns about economic impacts, limited participation in decision making, and lack of transparency in revenue allocation (Rosselló-Nadal & Sard, 2026; Zou et al., 2024). These tensions can affect policy implementation, accelerate environmental degradation, and undermine the ecological and development goals that protected areas (Castrejón & Defeo, 2023; Zou et al., 2024).

Our study examines whether and how revenue allocation can shift initial levels of public acceptability. Specifically, we test the hypothesis that aligning revenue allocation with stakeholders' preferences increases the support for conservation entrance fees. Exploiting an event study, we test this hypothesis by analyzing public acceptability before, shortly after (one-month), and long after (nine-months) an increase in conservation entrance fees in the Galapagos National Park. We conduct this study in the Galápagos, as it is a marine biodiversity hotspot of global importance and the first natural area declared a UNESCO World Heritage Site (World Heritage Centre, 2007). With annual visitor numbers approaching 300,000, a seven-fold increase since the 1990s, tourism growth represents a critical concern for the conservation of endangered and endemic species (Observatorio de Turismo de Galápagos, 2025). Efforts to increase entrance fees were delayed for five years due to local resistance, including demonstrations and legal disputes, linked to fears about potential economic impacts and dissatisfaction with revenue use[1]. The implementation of the fee

[1]Social backlash and legal disputes surrounding the increase of Galápagos entrance fees include local congressmen and municipalities challenging the policy change in court, leading to constitutional actions aimed at delaying or avoiding its enforcement (Mantuano, 2025, February 27). Street demonstrations against the fee increase were often shared on social media, while the research team documented one of these events during fieldwork. International media has also covered the change in Galápagos entrance fee since the first attempt in 2019 to its implementation in 2024 (The Guardian, 2024, August 15; The New York Times, 2019,

increase in August 2024 provides a unique opportunity to examine stakeholder responses across multiple horizons before and after policy change (Gobierno del Regimen Especial de Galapagos, 2019).

We adopt a sequential mixed-method approach. The first phase of our work consists of a qualitative study, which explores the way residents interpret entrance fees within broader views about tourism growth, institutional trust, distributive fairness, and use of conservation revenues. Building on these insights, the second phase of this study implements three survey waves and estimate mixed-effects regressions to test whether aligning revenue allocation with stakeholder preferences increases public support and how this effect evolves following policy implementation. Our findings are threefold. First, stakeholder opposition is shaped by perceived lack of transparency, concerns about reinforcing inequalities, and perceptions that conservation revenues do not address basic community needs. Second, although visitors are less sensitive to the specific use of revenues, directing funds toward publicly preferred areas substantially increases policy acceptability among residents. Third, while opposition among residents remains high and persistent in the baseline, the effect of revenue reallocation strengthens over time, progressively improving support for the policy change. We quantify the marginal effects of incorporating stakeholders preferences on the probability of supporting entrance fee policies and recommend participatory strategies to sustain policy support and strengthen protected area governance in the long-term.

Our research relates to the literature on public acceptability of environmental policies and tourist taxes, particularly entrance fees to protected areas. Social acceptability research examines public support for contested policies, identifying determinants such as emotions, perceived fairness, institutional trust, and policy effectiveness (Rodriguez-Sanchez et al., 2018; Wüstenhagen et al., 2007). However, most studies remain descriptive, focusing on determinants of public support rather than identifying mechanisms capable of shifting initial levels of policy acceptability. Moreover, this approach has been applied primarily to carbon taxation, climate policy, and congestion charges (Dolšak et al., 2020; Schuitema et al., 2010; Uyduranoglu & Ozturk, 2020). Importantly, these studies largely assume that perceived effectiveness increases support (Baranzini & Carattini, 2017; Kallbekken & Sælen, 2011), overlooking contexts in which effectiveness may generate opposition. In

September 26).

nature-based tourism settings, policies such as entrance fee increases may be perceived as effective in reducing visitor demand, which residents associate with economic losses. As a result, perceived effectiveness can reduce, rather than increase, public acceptability. We provide evidence on this.

In the case of tourist taxes, recent work suggests that perceptions of the use of collected revenues correlate with support for such taxes (Rosselló-Nadal & Sard, 2026), yet evidence of the effect of allocating revenues to publicly preferred areas on the levels of public acceptability remains scarce. Existing public acceptability studies also rarely capture dynamic changes in attitudes following policy implementation, particularly over longer time horizons when perceived costs and benefits are realized (Carattini et al., 2018; Mach et al., 2020). In parallel, research on entrance fees has largely focused on demand elasticity and estimations of willingness-to-pay among visitors, with limited attention to the perspectives of residents despite their relevance for participatory governance and effective policy implementation (Daly et al., 2015; King et al., 2025; Mach et al., 2020; Mmopelwa et al., 2007). Together, there is a lack of understanding of what affects the public acceptability of environmental policies, such as conservation entrance fees, and how acceptability evolves following the real implementation of policies (Schuitema et al., 2010; Zou et al., 2024). Building carefully on these gaps, we address two research questions: (1) Can allocating additional revenues from an entrance fee increase to publicly preferred areas shift initial levels of public acceptability? and (2) How does this effect change shortly after and long after policy implementation?

This study contributes in three ways. First, it conceptualizes and empirically tests revenue allocation as a mechanism that can shift initial levels of public acceptability of entrance fee policies. While prior research discusses associations between support and perceptions of revenue use (Rosselló-Nadal & Sard, 2026), we directly evidence that aligning revenue allocation with stakeholder preferences increases policy support and quantify its marginal effects. Second, this study advances a temporal perspective on public acceptability by examining how attitudes evolve following actual policy implementation across multiple time horizons. While Mach et al. (2020) analyze attitudes following the removal of entrance fees over a short period, our study evaluates acceptability of a substantial entrance fee increase across three stages: before, one month after, and nine months after implementation. Third, by integrating perceptions of tourism sustainability, distributive fairness, and conservation revenues with a quantitative analysis of stakeholder attitudes, the study provides a comprehensive understanding of the way entrance fee policies are evaluated and strategies to strengthen policy support through participatory and transparent revenue allocation. Beyond the Galápagos Islands, this study contributes to broader debates on conservation financing and tourism governance, offering recommendations to enhance the political feasibility of contested policies, aimed at facilitating their timely and effective implementation with lessons for the governance of other biodiversity-rich destinations.

The remainder of the paper is structured as follows. Section 2 presents the study area and the entrance fee policy. Section 3 describes the theoretical framework. Section 4 details the qualitative and quantitative research design, including empirical specification. Section 5 presents the thematic analysis and econometric findings. Section 6 discusses implications for public acceptability and protected area governance and concludes with directions for future research.

2 Study area and policy change

2.1 Study area

The Galápagos Islands, located in the Eastern Tropical Pacific Ocean approximately 900 km from continental Ecuador, are internationally renowned for their rich biodiversity, hosting more than 2,900 marine species with 18% endemism (World Heritage Centre, 2018). Following Charles Darwin's expedition aboard the *HMS Beagle* in 1835, whose observations inspired the formulation of the theory of evolution, the archipelago has been widely recognized as a natural laboratory of global scientific importance (Grant & Grant, 2008; Nature, 1935). Comprising 19 major islands and more than 100 islets and rocks, Galápagos is currently home to approximately 30,000 residents concentrated on four inhabited islands. Governance operates under the Special Regime Law for the Galápagos which imposes strict residency rules, regulates terrestrial and marine tourism, and establishes conservation entrance fees for visitors, from which residents are exempt (CGREG, 2015).

Designated a UNESCO World Heritage Site in 1978, the Galápagos were placed on the List of World Heritage in Danger in 2007 due to concerns related to invasive species, overfishing, and unsustainable tourism growth (World Heritage Centre, 2007). Although removed from the list in 2010, concerns about its rapid tourist expansion persist. In 2018, the Ecuadorian government committed to a "zero-growth" tourism model aimed at mitigating the increasing pressures of over-tourism (World Heritage Centre, 2018). However, tourism continues to expand: by 2024, annual visits reached 279,200, six times higher than in the 1990s and nearly ten times the resident population (Fig. 1). Domestic tourists accounted for 45% of total visits, while foreign visitors, mainly from the United States and the United Kingdom, accounted for the other 55% (Observatorio de Turismo de Galápagos, 2025).

Their global importance, strong tourism dependence, and persistent governance tensions make the Galápagos Islands a relevant setting for examining the political feasibility of key policy instruments, particularly entrance fees designed to balance conservation goals and local development.

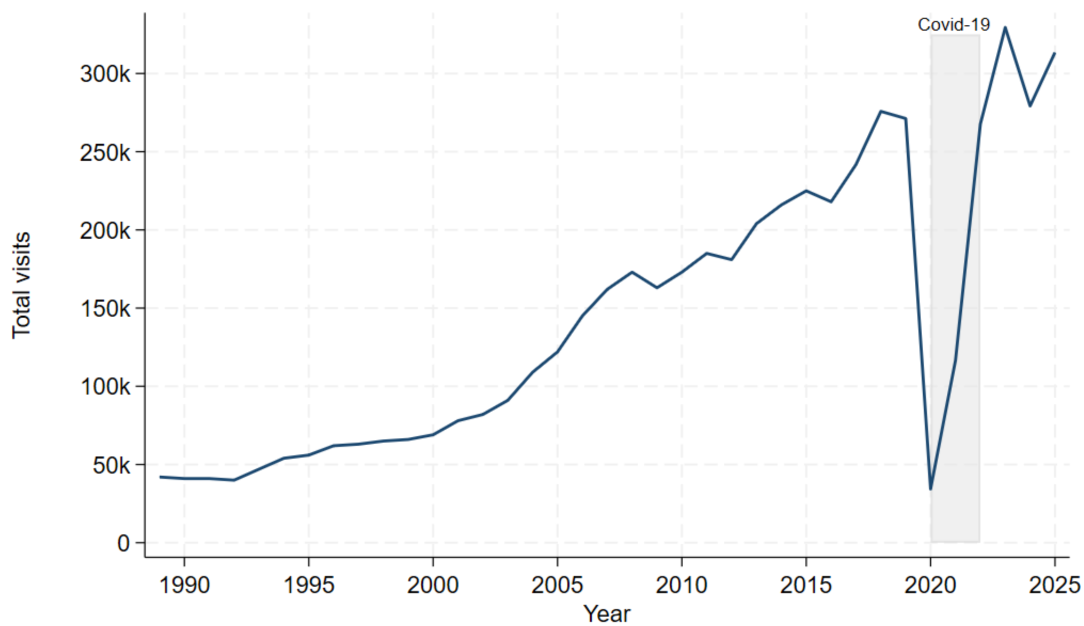

Figure 1: Number of tourists visiting the Galápagos National Park, 1990-2025
Source: Observatorio de Turismo de Galápagos (2025) and Pizzitutti et al. (2017)

2.2 Entrance fee policy

Entrance fees have been the main instrument to manage visitor flows and finance conservation in the Galápagos, but changes to the fee scheme have been highly contentious. Since their introduction in 1998, fees remained unchanged for 25 years with community resistance, demonstrations, legal disputes, and concerns over revenue use repeatedly delaying any changes. After a first attempt five years earlier, in February 2024, the authorities announced the entrance fee adjustment: fees increased from US$6 to US$30 for domestic visitors (a 400% increase) and from US$100 to US$200 for foreign visitors (a 100% increase), effective on August 1st 2024 despite persistent opposition among different stakeholders (Consejo de Gobierno del Régimen Especial de Galápagos, 2024; King et al., 2025).

Beyond regulating visits, the new fee scheme aims to generate resources to offset the effects of inflation since its introduction, which has risen the costs of managing and monitoring its terrestrial and marine protected areas. However, the long-term success of the new entrance fee scheme is more likely with public support. For a fragile ecosystem like the Galápagos, avoiding the sustainable tourism paradox, where tourism undermine the natural resources they are meant to protect, requires addressing conservation goals together with community livelihoods (Burbano et al., 2022).

3 Theoretical framework and empirical specification

We develop a theoretical framework, informed by the literature on public acceptability, environmental taxes, and visitor fees, to describe how individuals form their attitudes toward an entrance fee policy over time (Chung et al., 2011; Kallbekken & Sælen, 2011; Schuitema et al., 2010; Zhang et al., 2023).

Following Kallbekken and Sælen (2011) on environmental taxes, individuals evaluate a policy based on their beliefs about its consequences. Similarly, we assume that the individual's latent acceptability is shaped by the expected value derived from the policy change. In our setting, this includes two main elements: the expected costs and the expected benefits derived from the policy. For an entrance fee increase, expected benefits include improvements in ecological quality and public services resulting from additional government investment. Expected costs differ across stakeholder groups. For tourists, the cost is direct: the fee paid to access the protected area. For residents (who are exempt from paying the fee), costs are indirect and depend on the expected economic impact of changes in tourist demand. The expected value of the policy under scenario s at time t can be expressed as (Eq. 1):

$$\tilde{V}_{t,s} = \underbrace{v_p(p_{t,s}) + v_e(e_{t,s})}_{\text{Public Benefits (+)}} + \underbrace{v_c(c_{t,s})}_{\text{Private cost (-)}} \tag{1}$$

where $p_{t,s}$ and $e_{t,s}$ represent public services and ecological quality, respectively, and $c_{t,s}$ denotes private costs, such as the direct fee paid by tourists and the impact on tourist demand for residents. The value functions $v_p(\cdot)$ and $v_e(\cdot)$ are increasing and capture the policy benefits, while $v_c(\cdot)$ is decreasing and concave in costs.

The mechanism tested in this study is that the allocation of public revenues following individuals' preferences affects their expectations of benefits derived from the policy, consequently shifting the level of acceptability. Specifically, we hypothesize that directing additional revenues toward publicly preferred areas increases expected benefits (e.g., improved public services and ecological quality aligned with public needs) while holding private costs constant, i.e., the fee amount paid by visitors and the residents' expected impact on tourism demand remain unchanged. From here onward, we refer to the use of additional revenues in public's preferred areas as *reallocation*. As expected benefits increase in the revenue reallocation scenario ($s = 1$), the acceptability of the policy change is hypothesized to be higher than in the baseline scenario without reallocation ($s = 0$) (Eq. 2):

$$\uparrow v_p(p_{t,s=1}) + v_e(e_{t,s=1}) \Rightarrow Acceptability_{t,s=1} > Acceptability_{t,s=0} \tag{2}$$

Over time, as the policy is implemented and its outcomes become observable, perceptions of costs and benefits update, thereby altering the acceptability of the policy. We hypothesize the dynamics presented in Table 1, which describes changes in public acceptability for the combination of three policy change horizons and the two revenue allocation scenarios:

Table 1: Public acceptability dynamics: Policy change and revenue allocation

Policy change	Revenue allocation	
	No reallocation	*If reallocated*
Before policy	In the pre-implementation period where no changes in revenue allocation are proposed, acceptability is hypothesized to be low if *ex-ante* expected private costs outweigh expected public benefits, i.e., the potential costs dominates.	The allocation of revenues according to public's preferences ($s = 1$) shifts the balance of the value that is expected to be derived from the policy by raising the value of public benefits relative to private costs, thereby increasing acceptability compared to the baseline ($s = 0$).
Shortly after	In the immediate aftermath (t = *shortly-after*), policy outcomes are not fully realized. If initial expected costs were dominant, *ex-post* costs expectations are presumed to remain dominant in the short term, leading to persistently low acceptability.	Combining revenue reallocation ($s = 1$) with the policy implementation exposure (t = *shortly-after*) is expected to reinforce the value that individuals assign to the potential benefits derived from the policy if revenues are allocated according to their preferences; this leads to an amplified increase in acceptability.
Long after	Over a longer period (t = *long-after*), realized outcomes reinforce or attenuate initial expectations about policy outcomes. If observed economic impacts (*ex-post* perceptions) are weaker than expected, acceptability may rise in the baseline scenario.	If *ex-post* perceived costs are similar to *ex-ante* and *ex-post* expectations, reallocating revenues ($s = 1$) may further increase support as it becomes a compensation for realized losses, and the policy change is less likely to be reversed.

Residents' expectations about policy costs depend on their beliefs about the price elasticity of tourist demand: $\epsilon = -\frac{\partial Q}{\partial P} \cdot \frac{P}{Q}$, where Q represents the quantity demanded and P is the price of the environmental good, i.e., the fee to access a protected area (Rosselló-Nadal & Sard, 2026). If residents believe demand to be elastic ($\epsilon > 1$), they anticipate a disproportionate reduction in tourist arrivals following the entrance fee increase, and therefore expect a negative impact on their private income. Following this, perceptions of costs and benefits can vary with contextual and personal factors such as economic sector, income, environmental attitudes, and trust in public institutions (Qiu et al., 2023). We use these variables as controls in the econometric specifications.

4 Research design and methods

4.1 Mixed-method design

We exploit an event study generated by the increase in conservation entrance fees at the Galápagos National Park, implemented on August 1, 2024. This event provides an opportunity to examine how public acceptability evolves as stakeholders are exposed to an actual policy change.

We employ a sequential mixed-method design. First, we conducted a qualitative study through focus groups and semi-structured interviews, followed by an in-person survey whose data are used in the quantitative analysis. The qualitative phase uses thematic analysis to understand the way residents conceptualize the entrance fee policy, how they assess its potential costs and benefits, and the narratives around tourism and conservation that shape public acceptability (e.g., transparency, institutional trust). In the quantitative phase, we employ a within-subject survey design implemented across three waves among residents and visitors. Through econometric analysis, we formally test whether directing additional revenues toward publicly preferred areas increases acceptability and whether this effect changes shortly after and long after policy implementation compared with the pre-implementation period.

4.2 Qualitative study

For the qualitative study, we conducted six focus groups and six semi-structured interviews with 26 residents in Santa Cruz, Galápagos, in June 2024. Participants were purposively selected to represent sectors with distinct exposure to tourism and conservation governance, including tourism and hospitality, artisanal fishing, agriculture, NGOs, research, and local government. Sessions were audio-recorded and transcribed verbatim; when audio-recording was not possible, detailed notes were taken during and immediately after the session. Discussions were held in Spanish and transcripts were translated into English by bilingual team members.

We analyze qualitative data from these sessions using a hybrid inductive-deductive thematic analysis following the six phases proposed by Braun and Clarke (2006). Transcripts were first reviewed for familiarization and then open-coded to capture statements related to tourism sustainability, transparency, equity, and public concerns about revenue use. Across iterations, codes were compared, refined and clustered into analytical categories emerging from the data while being informed by the literature and our theoretical framework (Saldaña, 2013). We subsequently organized categories into four higher-order themes: (i) Tourist Demand and Sustainable Growth; (ii) Trust and Transparency; (iii) Fairness and Inequality; and (iv) Conservation Revenues and Community Needs. We developed a codebook with 29 analytical codes nested within these themes and applied it to the dataset. Quotes are reported using anonymized identifiers with role descriptors.

Insights from this phase informed the design of the survey instrument, including development of the baseline and revenue reallocation scenarios; list of revenue allocation areas used in the preference elicitation task; and identification and validation of key constructs included in the survey and econometric analysis (e.g., institutional trust, environmental attitudes, economic sector). The qualitative findings also guided the interpretation of quantitative results.

4.3 Quantitative study

We conducted on-site surveys in three waves using a within-subject design, which elicits multiple acceptability responses from the same individual across policy scenarios (Davidian & Giltinan, 2003; Fuster & Zafar, 2023). The survey questionnaire was drafted based on the qualitative findings and the literature, and pretested it with seven individuals, refining wording and improving the interpretability of the public acceptability section. We conducted a pilot survey with 26 residents and 22 tourists, leading to further adjustments to question wording.

The core section of the survey followed three steps. First, respondents evaluated the acceptability of the entrance fee increase under a baseline scenario i.e., without changes in the allocation of additional revenues. Second, respondents ranked nine potential investment areas for allocating additional funds. These categories reflect legally permissible uses of conservation entrance fees and were validated with protected area managers. To avoid order effects and mitigate anchoring bias, the order of items was randomized for each respondent (Johnston et al., 2017). Third, respondents reassessed the acceptability of the fee increase under a revenue reallocation scenario in which additional revenues are allocated according to their stated preferences. This design allows us to identify potential changes in acceptability when the policy is explicitly linked to public's preferred uses of revenues.

To study dynamics over time, we implemented the same questionnaire in three survey waves, which enables us to compare public acceptability of increasing conservation entrance fees in the short term (before vs one month after implementation) and in a longer horizon (a follow-up nine months after implementation). We selected a within-subject design due to its statistical efficiency relative to between-subject designs (Fuster & Zafar, 2023), this was important given the logistical difficulties of conducting fieldwork in the Galápagos and the event-study setting. The design is also policy-relevant, as most stakeholders were aware of the imminent (pre-policy) or recent (post-policy) fee increase.

The full questionnaire consisted of four sections. First, before starting the survey, respondents were informed about the research team, confidentiality protocols, and ethical approval. They were also informed that aggregated results would be shared with policymakers, a practice recommended to strengthen perceived consequentiality in stated preference research to encourage truthful responses (Carson & Groves, 2007). Second, after providing consent, respondents answered warm-up questions on recreational preferences, conservation perceptions, and prior knowledge of the planned or recent entrance fee change. Third, the main section elicited the acceptability of increasing conservation entrance fees using a five-point Likert scale ranging from "Strongly Oppose" to "Strongly

Favor", following Rodriguez-Sanchez et al. (2018). Residents were additionally asked about expected impacts of the fee increase on tourist visits and their trust in public institutions (measured on a five-star scale). Finally, respondents completed demographic questions, the New Ecological Paradigm (NEP) battery to measure environmental attitudes (Dunlap et al., 2000; Zhu & Lu, 2017), and an open-ended section for additional comments. In the case of tourists, they also reported travel characteristics, including number of previous visits, length of stay, travel experience, and visit modality (land-based or liveaboard).

We administered 1,358 in-person surveys to residents and visitors: 325 surveys were collected before the policy change (July 2024), 437 shortly after implementation (August–September 2024), and 596 nine months after implementation (May–June 2025). Data were collected by two trained local assistants under the supervision of the research team. Resident surveys ($n = 629$) were conducted face-to-face in Santa Cruz, Galápagos, using tablets and the SurveyLab platform. We used quota sampling to approximate the population distribution by gender and age. Given that roughly 60% of local employment is directly or indirectly linked to tourism (e.g., commerce, hospitality, transport), we also aimed to reflect this distribution in the sample (Pizzitutti et al., 2017). Tourist surveys ($n = 729$) were administered in paper format at the departure area of Baltra Airport, the main gateway to the archipelago, using systematic sampling. Field assistants approached travelers seated at every fifth seat; if the selected traveler refused, was under 18 years old, or was identified as a resident rather than a tourist, the next traveler was selected. Questionnaires were available in Spanish and English, and clarifications were provided when needed. Only participants aged 18 or older who gave informed consent were surveyed. Four tourist surveys were excluded from the econometric analysis because they were incomplete, while all resident surveys were complete and retained for analysis.

4.4 Empirical specification

We assess the hypothesized dynamics discussed in Table 1 by estimating latent acceptability $Acceptability^*_{s,i}$ as a function of exposure to policy implementation, revenue allocation, and their interaction; we also consider controls such as socio-economic and attitudinal variables, as shown in Eq. (3) and detailed below:

$$
\begin{aligned}
Acceptability^*_{is} = \delta_1 ShortlyAfter_i + \delta_2 Reallocation_s \\
+\delta_3 ShortlyAfter_i \times Reallocation_s + \beta X_i + \mu_i + \varepsilon_{is}
\end{aligned}
\tag{3}
$$

where $ShortlyAfter_{is}$ equals 1 if the acceptability of individual i was elicited shortly after policy implementation and 0 otherwise ($BeforePolicy$); $Reallocation_s$ equals 1 if the policy is presented in terms of allocating revenues according to public's preferences ($IfReallocated$), and 0 if presented under the baseline scenario ($NoReallocation$). The interaction term captures how the effect of reallocating revenues evolves in the short-term after policy implementation. X_i includes covariates such as demographics, trust in public institutions, and environmental attitudes. μ_i is a random intercept, accounting for acceptability being measured for the same individual under both revenue use scenarios, and $\epsilon_{s,i}$ is the error term.

To evaluate the dynamics of a longer post-implementation period, we extend the model by including the variable $LongAfter_i$, which equals 1 if the acceptability of individual i was elicited long after policy implementation and 0 otherwise; we also include its interaction with the variable $Reallocation_{s,i}$. The model is then defined by Eq. (4):

$$
\begin{aligned}
Acceptability^*_{is} = \delta_1 ShortlyAfter_i + \delta_2 LongAfter_i + \delta_3 Reallocation_s \\
+\delta_4 ShortlyAfter_i \times Reallocation_s + \delta_5 LongAfter_i \times Reallocation_s \\
+\beta X_i + \mu_i + \varepsilon_{is}
\end{aligned}
\tag{4}
$$

The observed dependent variable is measured on a five-point Likert scale ranging from Strongly Oppose (1) to Strongly Favor (5) as in Rodriguez-Sanchez et al. (2018), serving as an ordered proxy of the latent acceptability variable (Williams & Quiroz, 2020). Given the ordinal outcome and repeated-measures design, we estimate a mixed-effects ordered logit model (Davidian & Giltinan, 2003). Residents and tourists are analyzed separately to account for differences in the dynamics of their attitudes over time, as they are exposed to policy costs and policy implementation differently.

5 Results

5.1 Qualitative findings

The qualitative analysis, applied to recorded sessions with residents ($n = 26$), identified four higher-order themes that help explain how residents evaluate entrance fee increases and under what conditions they become acceptable: (i) Tourist Demand and Sustainable Growth; (ii) Trust and Transparency; (iii) Policy Fairness and Economic Inequality; and (iv) Conservation Revenues and Community Needs. Below, we use illustrative quotes to describe each theme and highlight how each maps onto the survey design and empirical specification.

5.1.1 Tourist Demand and Sustainable Growth

Debates about tourism growth in the Galápagos Islands reflect tensions between economic dependence and ecological sustainability (Almeida García et al., 2015). Residents interpret the entrance fee policies within broader narratives about the appropriate tourism model for the archipelago. For some participants, particularly those working in tourism, growth was portrayed as economically necessary and structurally difficult to reverse. Tourism was framed as the financial engine of conservation, reinforcing a mutually dependent relationship between environmental protection and increased visitor flows. A tour operator stated:

> "It was obvious that this [tourism growth] would happen. Tourism is needed to finance conservation and a good story always sells better. This brings in more people, requires more labor, and generates income to pay them. This will just continue" (Tour operator Q27).

From this perspective, entrance fees were viewed less as instruments to regulate tourism growth and more as mechanisms to finance its consequences. In contrast, participants from environmental and fishing sectors highlighted their concerns about exceeding the ecological carrying capacity of the islands and the environmental risks associated with greater visitor numbers. They interpreted fee increases through the lens of long-term sustainability, questioning whether the current tourism model can absorb continued growth without degrading fragile habitats. A tour operator also acknowledged that the economic benefits derived from tourism may not offset long-term environmental risks:

> "The increase in tourists is something positive. But there needs to be greater efforts on protecting the ecosystems we have, because if they are destroyed, the fall will later be drastic" (Tour operator Q33).

Residents also differentiated between visit modalities, particularly liveaboard and land-based tourism. Liveaboard tourism was often perceived as economically extractive and weakly embedded in the local economy, whereas land-based tourism was associated both with an important revenue stream for local businesses and with growing pressures due to mass tourism and less environmentally conscious visitors. About the potential impact of higher entrance fees on demand, most participants believed foreign demand to be relatively price inelastic, while domestic demand was considered more sensitive to price increases. Nevertheless, some residents suggested that the new fee might remain affordable for domestic visitors.

These narratives reveal that support for entrance fees is embedded within broader views on tourism growth and mediated by beliefs about demand elasticity. In the quantitative design, these mechanisms are captured through controls for economic sector and environmental attitudes, as well as through questions eliciting residents' expectations and perceptions of the entrance fee impacts on tourist demand. Although the fee scheme is defined by country of origin only, the tourist survey includes controls for both nationality and visit modality (land-based vs. liveaboard), allowing us to compare attitudes across different tourism segments.

5.1.2 Trust and Transparency

Across interviews and focus groups, institutional trust emerged as a condition for policy acceptability. Even among participants who recognized the need for additional conservation funding, support for the entrance fee increase was contingent upon trust in institutional capacity, transparency, and accountability. The importance of trust in public institutions is consistent with work in acceptability of other environmental fees such as water and waste disposal charges (Rodriguez-Sanchez et al., 2018; Zhang et al., 2023). Policymaking was several times described by residents as reactive and inconsistent, characterized by “patchwork solutions” and decisions made “hastily.” It is interesting to note that some residents questioned whether the fee increase was primarily motivated by fiscal needs or international reputation rather than sustainability objectives or local priorities:

> “The decision to increase the fee was made solely as a strategy to collect more funds, and not necessarily to limit tourism, since no [visitor] caps have been established and the new fee will not restrict the growth in visitor numbers.” (Marine biologist Q55).

Concerns about conservation being instrumentalized for tourism promotion further weakened trust in institutions, with skepticism about the motivations behind the policy change and whether revenues would be managed effectively. Transparency in revenue allocation repeatedly emerged as a factor in shaping support:

> ”I would lean more in favor of the increase if there were clarity and certainty about where the funds will go” (Hospitality entrepreneur Q34).

These narratives suggest that residents evaluate entrance fee policies based both on expected benefits and procedural legitimacy. Trust shapes expectations about whether potential benefits will materialize and whether revenues will be managed transparently. This mechanism directly informed the quantitative design. The questionnare measures trust in public institutions and elicits perceived survey consequentiality, allowing us to test and quantify the degree to which institutional trust explains the acceptability of increasing entrance fees.

5.1.3 Policy Fairness and Economic Inequality

Perceptions of distributive injustice and unfairness within the tourism economy also emerged as key narratives through which residents assess the entrance fee policy. Tourism growth was frequently described as producing asymmetric benefits, with local businesses generating limited profits relative to large tourism operators. Expressions such as “We are only left with the crumbs” (NGO member, Q14) captured concerns about economic inequality. Liveaboard tourism was described as highly profitable yet contributing less to the local economy than land-based tourism. Hospitality entrepreneurs linked this dynamic to intensified competition among small businesses and declining margins:

> "Backpackers generate more income to local businesses [than liveaboard passengers]. We are forced to compete fiercely among ourselves. It's like a war. This competition is getting out of hand" (Restaurant owner Q31).

Concerns about fiscal centralization were also frequent. Participants noted that collected revenues are first transferred to the national treasury before being redistributed to local institutions, raising fears that such funds may not fully return to the islands if not executed in an efficient and timely manner. These concerns are consistent with official figures showing that between 2015 and 2023 the collection of entrance fee revenues increased while public spending by the national park authority declined (Consejo de Gobierno del Régimen Especial de Galápagos, 2024).

In short, residents also evaluated entrance fees through distributive justice considerations: who pays, who benefits, and whether the distribution of costs and benefits is perceived as fair; consistent with previous studies on attitudes toward protected area fees (Mach et al., 2020). When the policy was perceived as reinforcing existing inequalities, its legitimacy among residents weakened. These insights informed the quantitative strategy in two ways. First, the econometric specifications control for income and employment sector to capture differential exposure to tourism benefits. Second, the revenue reallocation scenario directly tests whether redistribution toward publicly preferred areas can increase policy acceptability.

5.1.4 Conservation Revenues and Community Needs

Finally, residents assessed the entrance fee policy through its capacity to deliver tangible improvements in everyday life. While conservation funding needs were acknowledged, many participants emphasized deficits in basic services, including as more immediate priorities: safe water, sanitation, waste management, and infrastructure. A dive guide expressed:

> "Health comes first, we need better public services like safe water and sewage systems, where we are currently stagnant. Then infrastructure, not only for visitor sites but also for docks and ports" (Scuba diving leader Q25).

For many, conservation and social investment were complementary rather than competing goals. However, social needs were often framed as a precondition for strengthening environmental awareness and reducing tensions between residents and conservation authorities. A marine biologist noted:

> "The new fees [scheme] will allow for the collection of more funds that are necessary for different investments, but these should first address the needs of the population. If people's needs are not met, they will not be able to think about conservation" (Marine biologist Q50).

Overall, these narratives suggest that residents evaluate entrance fees largely through their capacity to deliver tangible benefits; policy legitimacy depended on whether revenues would translate into visible improvements and internalize tourism environmental externalities. This theme directly informed the quantitative study design. The list of potential revenue allocation areas included in the survey was derived from these discussions and subsequently validated with authorities to ensure compliance with legally permissible uses.

5.2 Quantitative results

5.2.1 Sample characteristics

Table 2 reports descriptive statistics for the resident sample ($n = 629$) across the three survey waves: before policy, one month after and nine months after policy implementation. Gender distribution is balanced with 46-53% males and 47-54% females. Respondents range from 18 to 70 years old, with mean ages between 35 and 37. Average monthly income is approximately USD 970, comparable to the 2021 Galápagos Census figure (USD 978) (INEC, 2023, 2024). In terms of education, 44–63% completed secondary education or lower, and 37–56% hold undergraduate or postgraduate degrees. Employment profiles reflect the tourism-based economy: 44–50% work directly in tourism (e.g., tour agencies, hospitality, transport), 15–19% in commerce, 10–12% in government or conservation (including NGOs and research), and 7–11% are students or unemployed. Most residents reported being aware of the policy change. Statistical tests indicate no differences in age across waves, while income differs across at least one wave at the 5% level; gender and economic sector are not statistically different between waves. For comparability in the econometric analysis, we use entropy balancing weights to match key characteristics across survey waves, including gender, age, education level, economic sector, and income. Balanced sample characteristics are reported in the Appendix (Table A.1).

Table 2: Sample descriptive characteristics - Local Residents

Variable		Before policy change	After policy one-month	After policy nine-month	Ho: No difference p-value
Residence	Non-permanent	14%	12%	10%	0.355
Status	Permanent	86%	88%	90%	
Gender	Male	51%	46%	53%	0.287
	Female	49%	54%	47%	
Age	Mean	37.4	36.3	34.8	0.062
18 - 70 years	(std. dev.)	(13.5)	(10.6)	(10.8)	
Education	Secondary or lower	55%	44%	63%	<0.001
level	Higher education	45%	56%	37%	
Monthly income	Mean	1.09k	0.91k	0.91k	0.019
USD 0.25k - 6k	(std. dev.)	(0.97)	(0.65)	(0.50)	
Economic	Commerce	15%	16%	19%	0.501
sector	Tourism-related	46%	50%	44%	0.406
	Govt. or conservation	10%	12%	12%	0.871
	Other sectors	23%	17%	14%	0.074
	Students and none	6%	5%	11%	0.026
Policy change	Yes	97%	99%	89%	–
awareness	No	3%	1%	11%	
Observations	–	203	210	216	N = 629

Notes: Survey waves are compared using analysis of variance (ANOVA). The null hypothesis (Ho) states that there are not statistical differences across all group means.

Table 3 summarizes tourist sample characteristics ($n = 729$). Across waves, the sample is split between 42-61% domestic visitors and 39-58% foreign visitors. Tourists range from 18 to 80 years old, with mean ages between 39 and 43. In terms of visit modality, 67–79% stayed land-based and 21–33% traveled on liveaboard cruises. No statistical differences are detected across waves for gender, income, occupation, or length of stay. The final wave includes relatively more foreign and highly educated tourists and a lower share of land-based visitors; tourists interviewed after implementation were also more aware of the entrance fee increase. For econometric analysis, we also apply entropy balancing to match demographics and visit modality (land-based or liveaboard) across waves. Balanced tourist characteristics are reported in the Appendix (Table A.2).

Table 3: Sample descriptive characteristics - Tourists

Variable		Before policy change	After policy one-month	After policy nine-months	Ho: No difference p-value
Tourist nationality	Domestic	61%	57%	48%	0.013
	Foreign	39%	43%	52%	
Gender	Male	41%	47%	50%	0.179
	Female	59%	53%	50%	
Age 18 - 83 years	Mean	42.7	38.8	41.1	0.034
	(std. dev.)	(15.1)	(14.4)	(13.7)	
Education	Secondary or lower	23.3%	16.3%	11.8%	0.008
	Higher education	76.7%	83.7%	88.2%	
Monthly income USD 0.3k - 150k	Mean	12.5k	8.3k	10.5k	0.388
	(std. dev.)	(33.0k)	(23.2k)	(27.2k)	
Occupation	Private worker	56%	56%	62%	0.323
	Public employee	25%	22%	24%	0.766
	Retired	11%	7%	7%	0.382
	Other	8%	15%	7%	0.010
Tourist modality	Land-based	79%	73%	67%	0.022
	Liveaboard	21%	27%	33%	
Lenght of stay 3 - 60 days	Mean	7.8	6.9	7.3	0.461
	(std. dev.)	(9.7)	(5.2)	(6.2)	
Policy change awareness	Yes	48%	62%	38%	<0.001
	No	52%	38%	62%	
Observations	–	122	227	380	N = 729

Notes: Survey waves are compared using analysis of variance (ANOVA). The null hypothesis (Ho) states that all group means are statistically equal.

5.2.2 Descriptive results

In the questionnaire, respondents were asked to indicate, on a 5-point Likert scale, their level of acceptability of increasing entrance fees to the Galápagos National Park for domestic and foreign visitors. Table 4 reports residents' acceptability under the baseline scenario (no revenue reallocation). Results show that residents' support is low: only 7% and 10% of residents strongly support increasing entrance fees for national and foreign tourists, respectively, while 17% and 14% remain indifferent. By contrast, around 51% strongly oppose increases for national tourists and 49% strongly oppose increases for foreign tourists. A Pearson's chi-squared test for differences in frequencies indicates that residents do not significantly differentiate between increases in entrance fees for national and foreign visitors.

Table 4: Residents' public acceptability

Public acceptability 5-point Likert scale	National fee increase	Foreign fee increase	Difference %	Pearson's chi-squared Ho: No difference
Strongly oppose	51%	49%	-2%	chi2 = 6.977
Somewhat oppose	16%	15%	-1%	p = 0.137
Neutral	17%	14%	-3%	
Somewhat favor	9%	12%	3%	
Strongly favor	7%	10%	3%	
Observations	629	629	-	-

Table 5 summarizes tourists' responses. In contrast to residents, the attitudes of tourists are more evenly distributed. We find that 15% and 13% of respondents strongly oppose fee increases for national and foreign visitors, respectively, while 28% and 21% remain indifferent, and 21% and 27% strongly support the increases. Overall, tourists exhibit greater acceptability of higher entrance fees than residents and, unlike residents, express stronger support for higher fees for foreign visitors.

Table 5: Tourists' public acceptability

Public acceptability 5-point Likert scale	National fee increase	Foreign fee increase	Difference %	Pearson's chi-squared Ho: No difference
Strongly oppose	15%	13%	-2%	chi2 = 15.685
Somewhat oppose	17%	17%	0%	p = 0.003
Neutral	28%	21%	7%	
Somewhat favor	19%	22%	3%	
Strongly favor	21%	27%	6%	
Observations	725	728	-	-

Respondents were also asked to rank their preferences for the use of additional revenues generated by the entrance fee increase by ordering investment areas from a list provided in the survey. Rankings were aggregated using the Rank Order Centroid method (Table 6). Among residents, the highest priority is investment in safe water and wastewater systems, followed by species conservation, improvements to tourist sites, infrastructure in populated areas, and solid waste management. Tourists, by contrast, place the highest priority on species conservation, followed by solid waste management, safe water and wastewater systems, protected area monitoring, and invasive species control. These rankings define the revenue reallocation scenario used later in the survey and reflect the importance placed on tangible policy benefits identified in the qualitative analysis, particularly among residents.

Table 6: Preferences for allocating additional revenues from the entrance fee increase - Rank Order Centroid (ROC) Method - Residents and Tourists

Residents			Tourists		
Rank	Areas for revenue use	Score	Rank	Areas for revenue use	Score
1st	Water and wastewater system	84.6	1st	Species conservation	122.1
2nd	Species conservation	78.3	2nd	Solid waste management	90.3
3rd	Improvement of tourist sites	65.2	3rd	Water and wastewater system	80.7
4th	Infrastructure in populated areas	64.2	4th	Protected areas management and monitoring	73.7
5th	Solid waste management	60.6	5th	Control of invasive species	73.3

Figures 2 and 3 illustrate how residents' public acceptability changes across survey waves and revenue use scenarios. For national fees (Fig. 2), 48% of residents strongly opposed the increase before implementation under the baseline scenario. However, when revenues are allocated to residents' preferred areas, strong opposition fell substantially to 29%, with more than one-third of initial opponents shifting toward support. This shift becomes more pronounced after policy implementation. Under the reallocation scenario, opposition declines further to 17% shortly after implementation and to 5% long after policy implementation. In contrast, opposition remains high under the baseline scenario (48% after one month) and increases to 56% nine months after implementation. Similar patterns are observed for residents' acceptability of increasing fees for foreign visitors (Fig. 3). Overall, these results suggest that residents' willingness to support the policy is strongly influenced by whether revenues are allocated in line with public preferences and priorities.

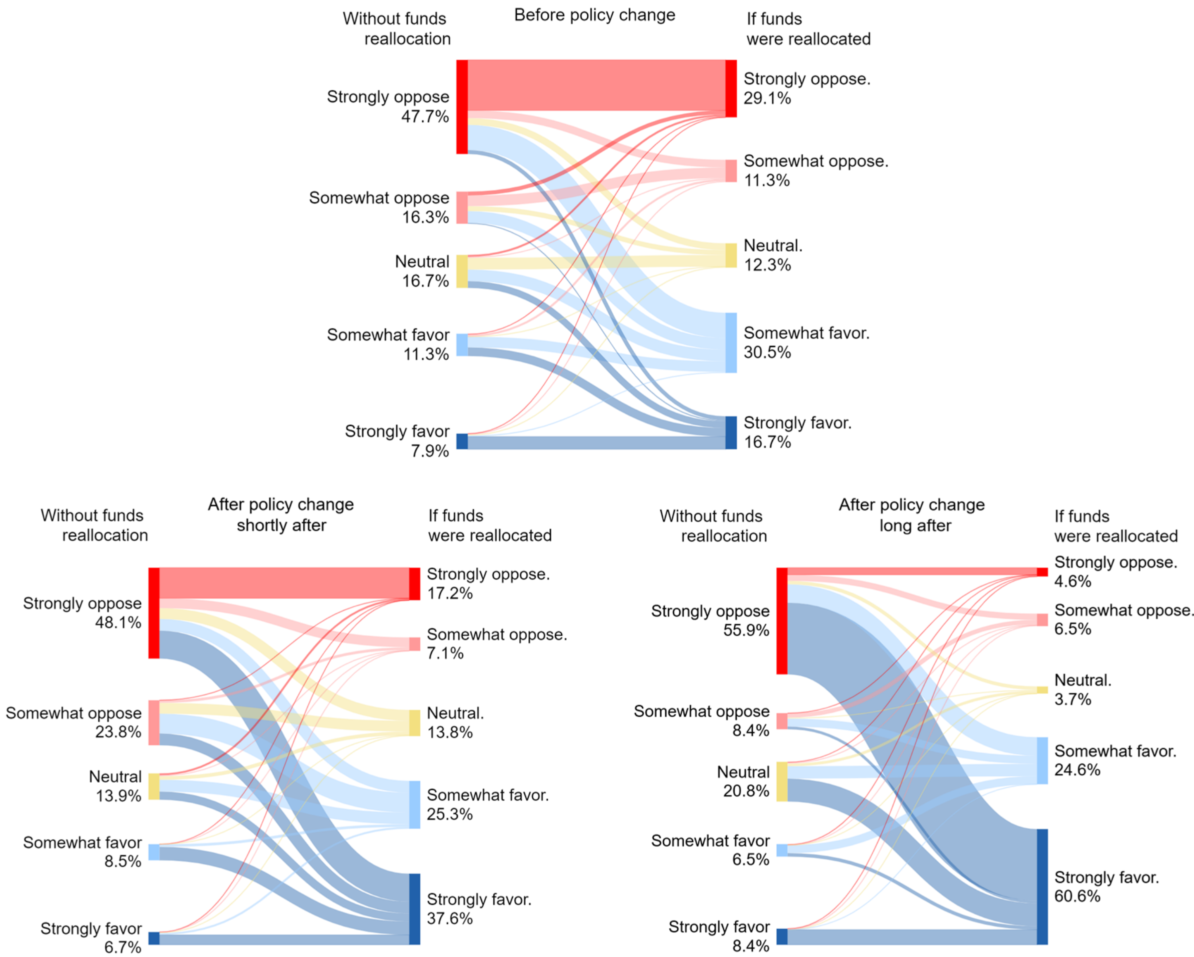

Figure 2: Residents' Public acceptability - National fee increase - By policy change period and revenues allocation scenario.

Notes: This figure shows residents' acceptability of increasing entrance fees to the Galápagos National Park for domestic visitors, before, shortly after (one-month), and long after (nine-month) policy implementation.

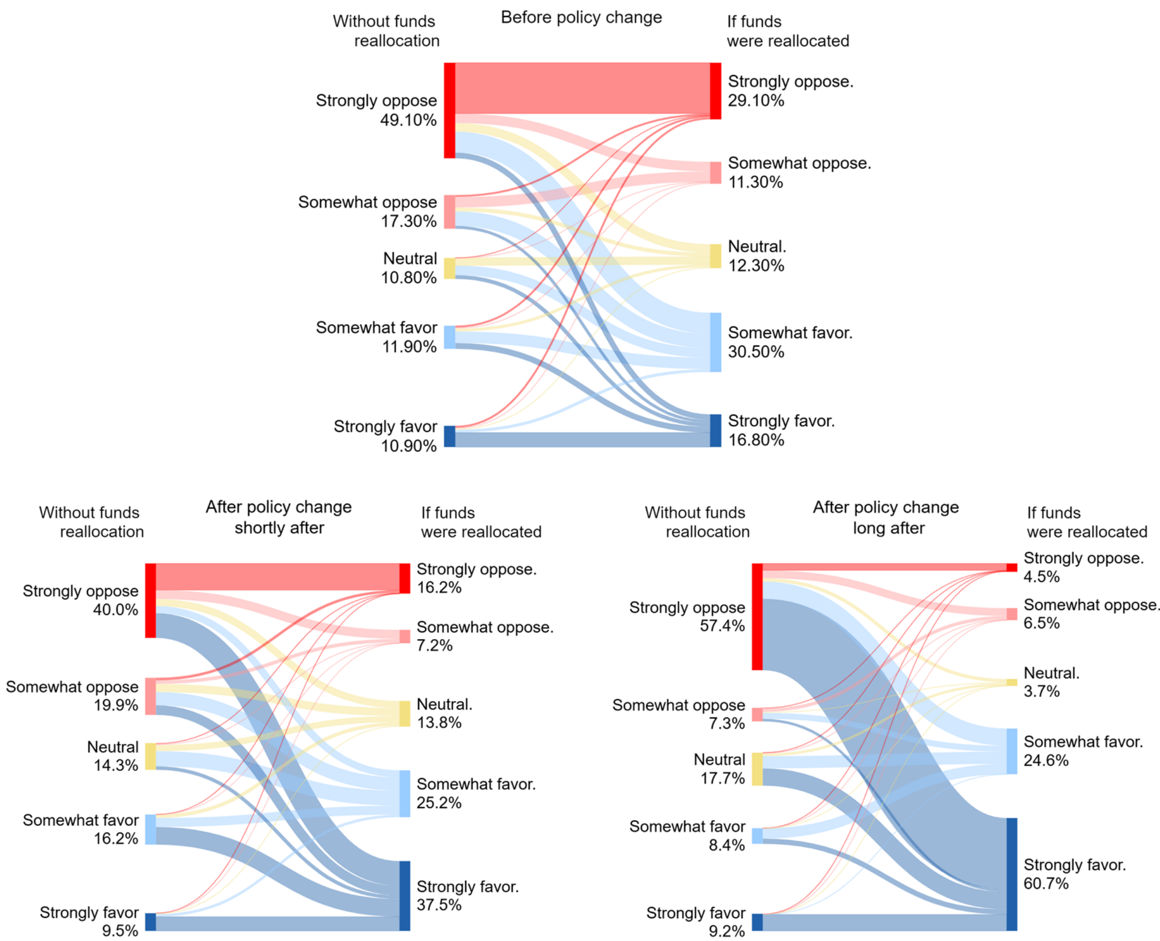

Figure 3: Residents' Public acceptability - Foreign fee increase - By policy change period and revenues allocation scenario.

Notes: This figure shows residents' acceptability of increasing entrance fees to the Galápagos National Park for foreign visitors, before, shortly after (one-month), and long after (nine-month) policy implementation.

A similar analysis is conducted for tourists, distinguishing between domestic (Ecuadorian) and foreign visitors, results are presented in Figures 4 and 5. Domestic tourists expressed stronger opposition before policy implementation, but this softens after enforcement, with responses shifting toward neutrality (Fig. 4). Foreign tourists remain consistently neutral or supportive (over 70%) of higher entrance fees both before and after implementation, under both revenue use scenarios, i.e., with and without reallocation (Fig. 5). When comparing scenarios, domestic tourists show a clearer positive shift in support under the reallocation scenario, whereas foreign tourists (already largely supportive) exhibit only modest changes. Additional statistical tests for residents and tourists are reported in the Appendix (Tables A.3-A.7).

Figure 4: Domestic Tourists' Public acceptability - National fee increase - By policy change period and revenues allocation scenario.

Notes: This figure shows domestic tourists' acceptability of increasing the entrance fee they pay to access the Galápagos National Park, before, shortly after (one-month), and long after (nine-month) policy implementation.

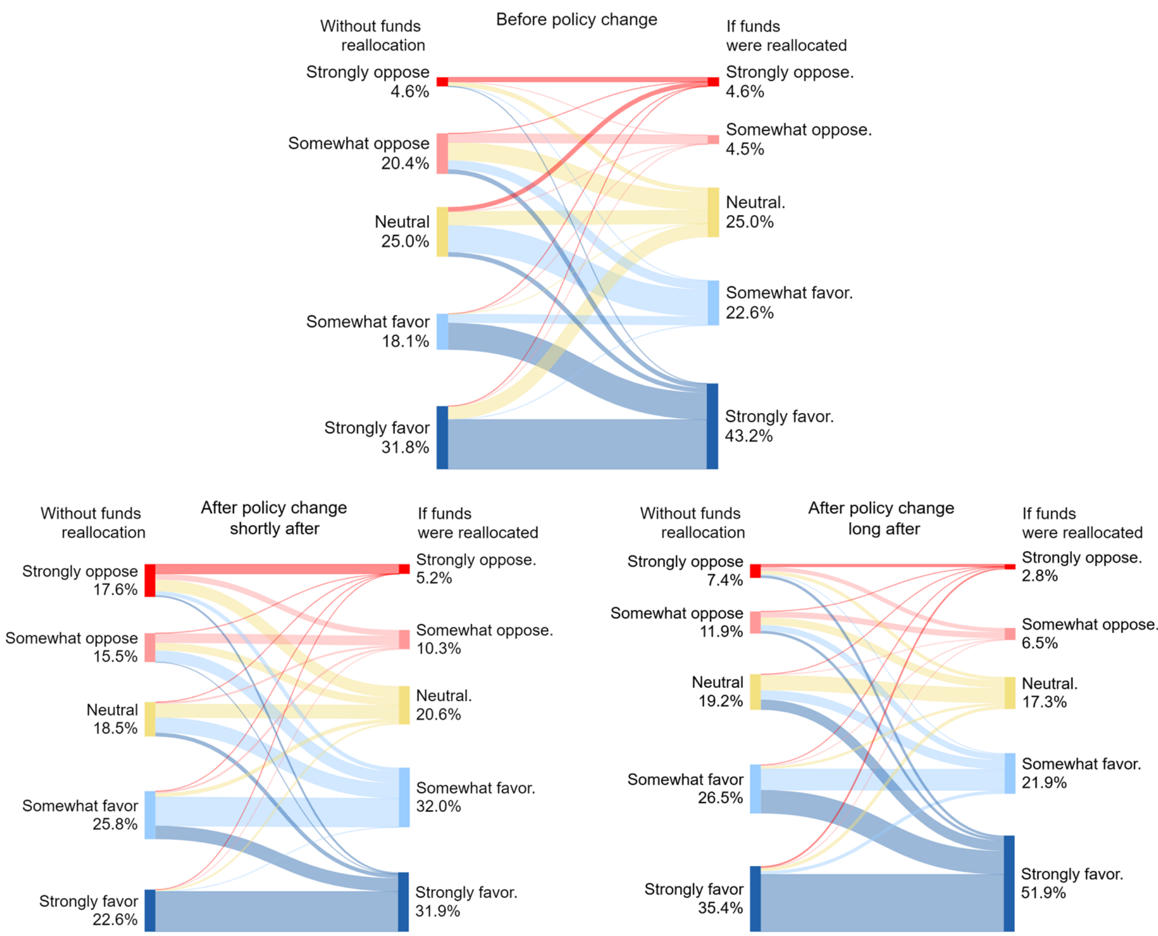

Figure 5: Foreign Tourists' Public acceptability - Foreign fee increase - By policy change period and revenues allocation scenario.

Notes: This figure shows foreign tourists' acceptability of increasing the entrance fee they pay to access the Galápagos National Park, before, shortly after (one-month), and long after (nine-month) policy implementation.

5.2.3 Econometric results: shortly after

We estimate mixed-effects ordered logit models of public acceptability following Eq. (3). Acceptability, measured on a five-point Likert scale, is modeled as a function of the key variables used to test the hypothesis: the variable *Shortly After*, indicating whether the policy change has recently been implemented; the scenario *If Reallocated*, which modifies the allocation of additional revenues; and their interaction. The baseline category for comparison is *Before Policy, No Reallocation.* To facilitate interpretation, Figures 6 and 7 present estimated coefficients and 95% confidence intervals for residents and tourists, respectively, across four specifications. Specifications (1)-(2) assess the acceptability of increasing entrance fees for national visitors, while specifications (3)-(4) assess increases for foreign tourists. Specifications (2) and (4) additionally include control variables such as demographic characteristics, trust in public institutions, environmental attitudes, and perceived survey consequentiality. Full regression tables are reported in the Appendix (Tables A.8-A.11).

For residents (Fig. 6), results show that under the baseline scenario, public acceptability is not statistically different after implementation compared to the period before policy change. However, revenue reallocation significantly increases acceptability, suggesting that directing revenues toward publicly preferred areas enhances expected policy benefits. Moreover, the interaction term is positive and significant, indicating that shortly after policy implementation the increase in public acceptability led by revenue reallocation is stronger than in the period before the policy took effect. According to specifications (2) and (4) in Figure 6, which include controls, trust in public institutions is positively associated with residents' support for increasing entrance fees for both national and foreign visitors. Stronger pro-environmental attitudes, higher income, and employment in government or conservation sectors are also positively associated with public acceptability. Coefficients for age, gender, and education are not statistically different from zero. These results align with the qualitative evidence indicating that policy support depends on perceived institutional credibility and tangible public benefits. The estimations are robust to models without entropy balancing weights.

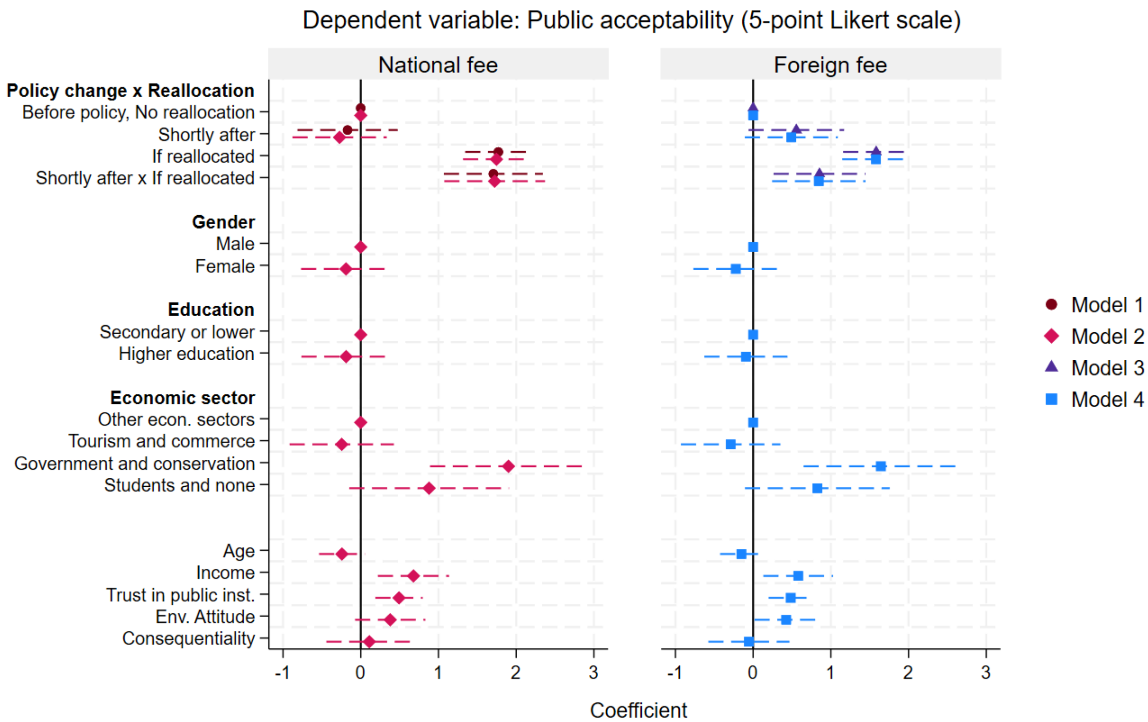

Figure 6: Residents results - Estimates of public acceptability before and shortly after policy change

Notes: This figure presents estimated coefficients and 95% confidence intervals (dashed lines) from mixed-effects ordered logit models of residents' support for increasing entrance fees to the Galápagos National Park, before and shortly after (one-month) policy implementation. Models 1–2 refer to support for increasing fees for domestic tourists, while Models 3–4 refer to fees for foreign tourists. Models 2 and 4 additionally controls for demographics, trust in public institutions, environmental attitudes, and perceived survey consequentiality. Regression tables are reported in the Appendix, Table A.8.

For tourists (Fig. 7), we find that reallocating revenues toward publicly preferred areas also increases public acceptability, although to a lesser extent as the magnitude of the coefficients is smaller and estimated with wider confidence intervals compared with residents. In contrast to residents, no significant differences are observed between the periods before and shortly after policy implementation under either revenue scenario. When control variables are included (Model 2 and 4), estimates show that domestic land-based tourists are consistently less supportive than liveaboard tourists and foreign visitors. Higher education and stronger pro-environmental attitudes are positively associated with public acceptability. Perceived survey consequentiality significantly increases tourists' support for the fee increase, whereas it has no significant effect among residents. Age, gender, and income are not significantly associated with acceptability, likely because income effects are captured by controls for nationality, visit modality, and education. Overall, these results indicate that aligning revenue allocation with stakeholder preferences increases short-term policy acceptability, especially among residents.

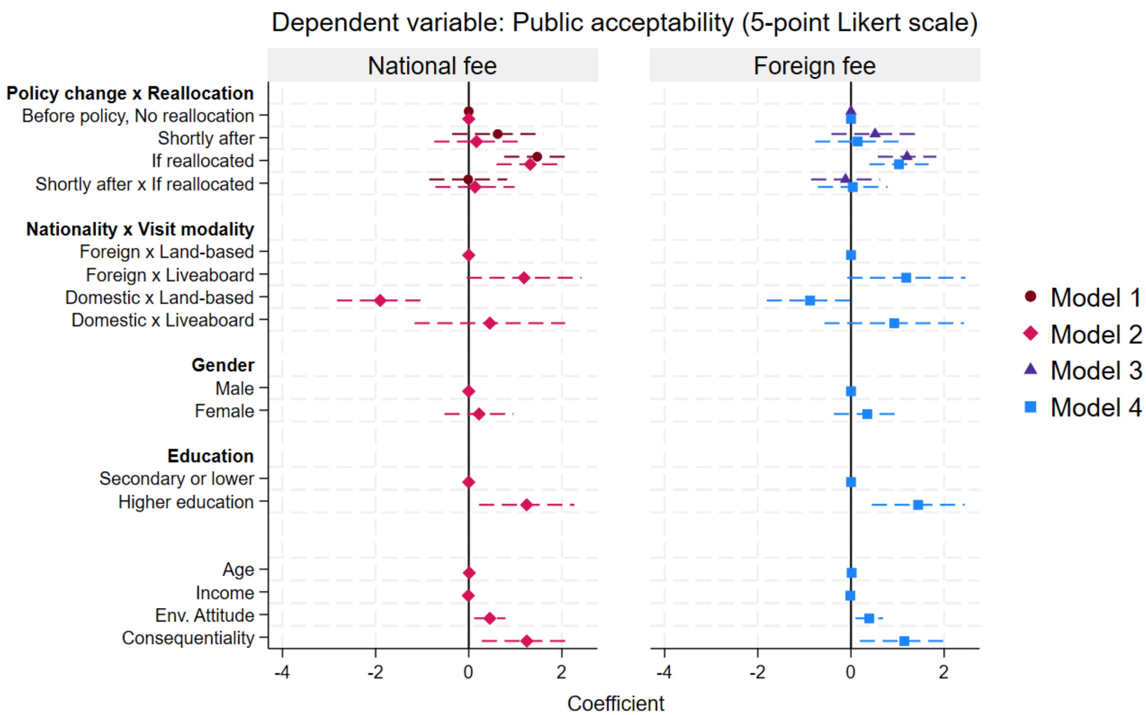

Figure 7: Tourists results - Estimates of public acceptability before and shortly after policy change

Notes: This figure presents estimated coefficients and 95% confidence intervals (dashed lines) from mixed-effects ordered logit models of tourists' support for increasing entrance fees to the Galápagos National Park, before and shortly after (one-month) policy implementation. Models 1–2 refer to support for increasing fees for domestic tourists, while Models 3–4 refer to fees for foreign tourists. Models 2 and 4 controls for demographics, visit modality, environmental attitudes, and perceived survey consequentiality. Regression tables are reported in the Appendix, Table A.9.

5.2.4 Econometric results: long after

We examine the longer-term dynamics of public acceptability following Equation (4), which incorporates the variable *Long After* policy change and its interaction with revenue reallocation. For residents (Fig. 8), results indicate that public acceptability slightly declined under the baseline scenario long after policy implementation (Models 1 and 3), although this change is not significant in specifications including covariates (Models 2 and 4). The persistence of low acceptability among residents is consistent with respondents perceiving realized policy costs as comparable to or greater than expected. Survey responses support this interpretation: before implementation, 85% of residents expected a reduction in tourist visits; nine months later, 72% reported perceiving fewer tourists, and two-thirds of this group attributed the decline directly to the entrance fee increase (Table 7).

In contrast, the marginal effect of revenue reallocation strengthens significantly over time. Figure 9 illustrates the marginal effects of the interaction terms across the five levels of public acceptability. Compared with the baseline scenario, reallocating revenues according to public preferences increases the probability of "Strongly Favoring" the entrance fee policy by approximately 10% before implementation, 30% shortly after, and 50% long after implementation. At the same time, revenue reallocation reduces the probability of "Strong Opposition" across these three periods. Together, these results suggest that residents' support increasingly depends on whether revenues are directed toward publicly valued benefits. These estimates are also robust to models without entropy balancing weights.

Table 7: Residents' expectations and perceptions of the entrance fee increase impact on tourist demand

	Ex-ante expectations Before policy change	Ex-post perceptions Long after policy change
Higher or equal number of tourists	15%	28%
Lower number of tourists	85%	72%
If responded "lower number of tourists"	Before policy change	Long after policy change
Due to the entry fee increase	–	68%
Due to conditions in the mainland	–	29%
Due to other reasons	–	3%

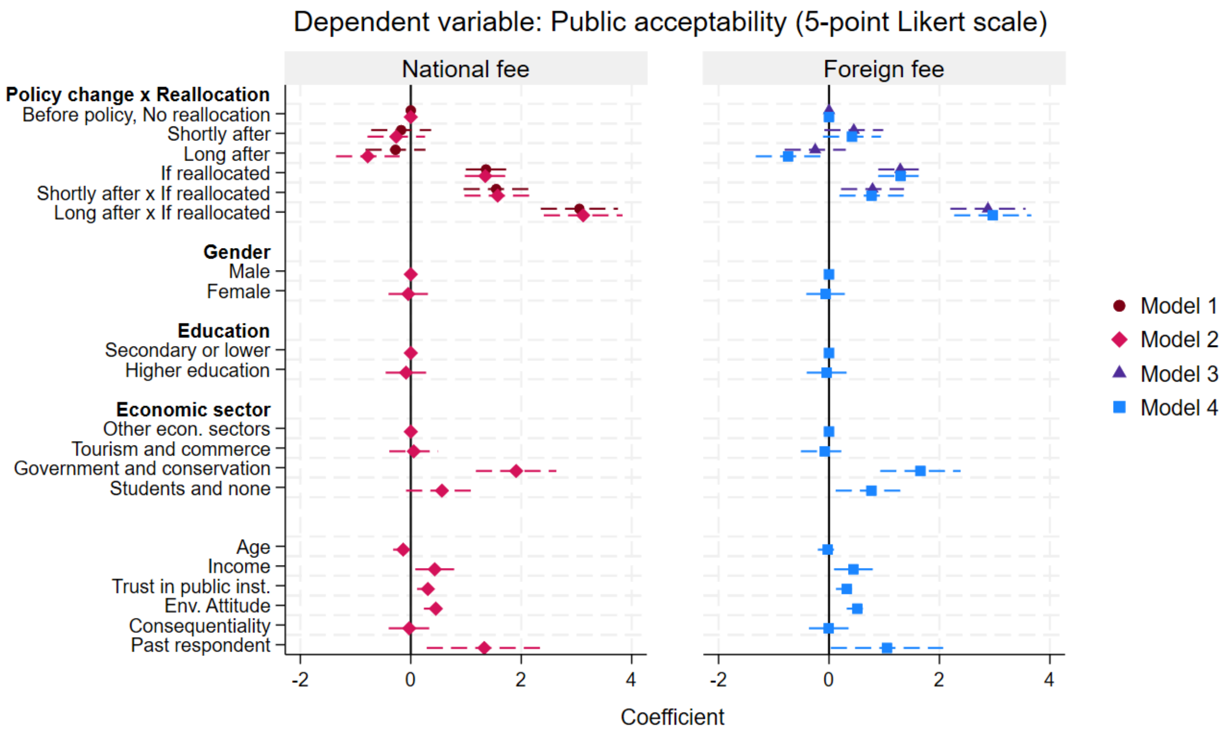

Figure 8: Residents results - Estimates of public acceptability long after policy change.
Notes: This figure presents estimated coefficients and 95% confidence intervals (dashed lines) from mixed-effects ordered logit models of residents' support for increasing entrance fees to the Galápagos National Park, before, shortly after (one-month) and long after (nine-month) policy implementation. Models 1–2 refer to support for increasing fees for domestic tourists, while Models 3–4 refer to fees for foreign tourists. Models 2 and 4 controls for demographics, trust in public institutions, environmental attitudes, and perceived survey consequentiality. Regression tables are reported in the Appendix, Table A.10.

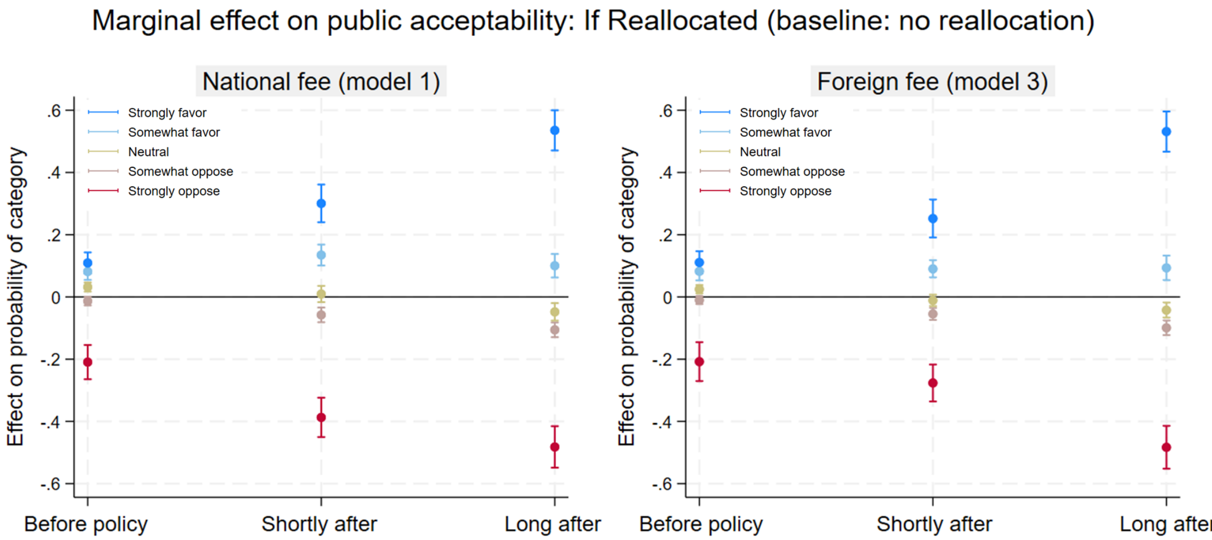

Figure 9: Residents results - Marginal effects of revenue reallocation on public acceptability by policy change period.

Notes: This figure reports the marginal effect of the revenue reallocation on residents' acceptability of increasing entrance fees to the Galápagos National Park, compared to the baseline scenario (no reallocation) across three periods: before, shortly after (one-month) and long after (nine-month) policy implementation. Estimates are based on Models 1 and 3, corresponding to increases in fees for domestic and foreign tourists, respectively. Regression tables are reported in the Appendix, Table A.10.

Figure 10 presents public acceptability estimates for tourists across the three survey waves. Under the baseline scenario, tourists' support for higher entrance fees appears to increase long after policy implementation (Models 1 and 3), although this change is not statistically significant once control variables are included (Models 2 and 4). The coefficient of revenue reallocation remains positive and statistically significant across specifications. However, the interaction with policy change shows that revenue reallocation is statistically significant only for increases in fees for national tourists long after policy implementation (Models 1 and 2).

Figure 11 illustrates the marginal effects of revenue reallocation across levels of public acceptability and policy change periods for tourists. Results show that long after implementation, reallocating revenues increases the probability of tourists "Strongly Favoring" higher entrance fees for national visitors by almost 20%. This increase is primarily driven by a reduction in the probability of "Somewhat Supporting" the policy, with no significant changes observed in the probabilities of neutrality or opposition (Model 1). For increasing fees to foreign visitors, the marginal effect of revenue reallocation remains relatively stable shortly and long after policy implementation (Model 3). Covariate patterns remain consistent with the short-term analysis: land-based tourists are less supportive than liveaboard tourists, while higher education, stronger environmental attitudes, and greater perceived survey consequentiality are associated with higher public support.

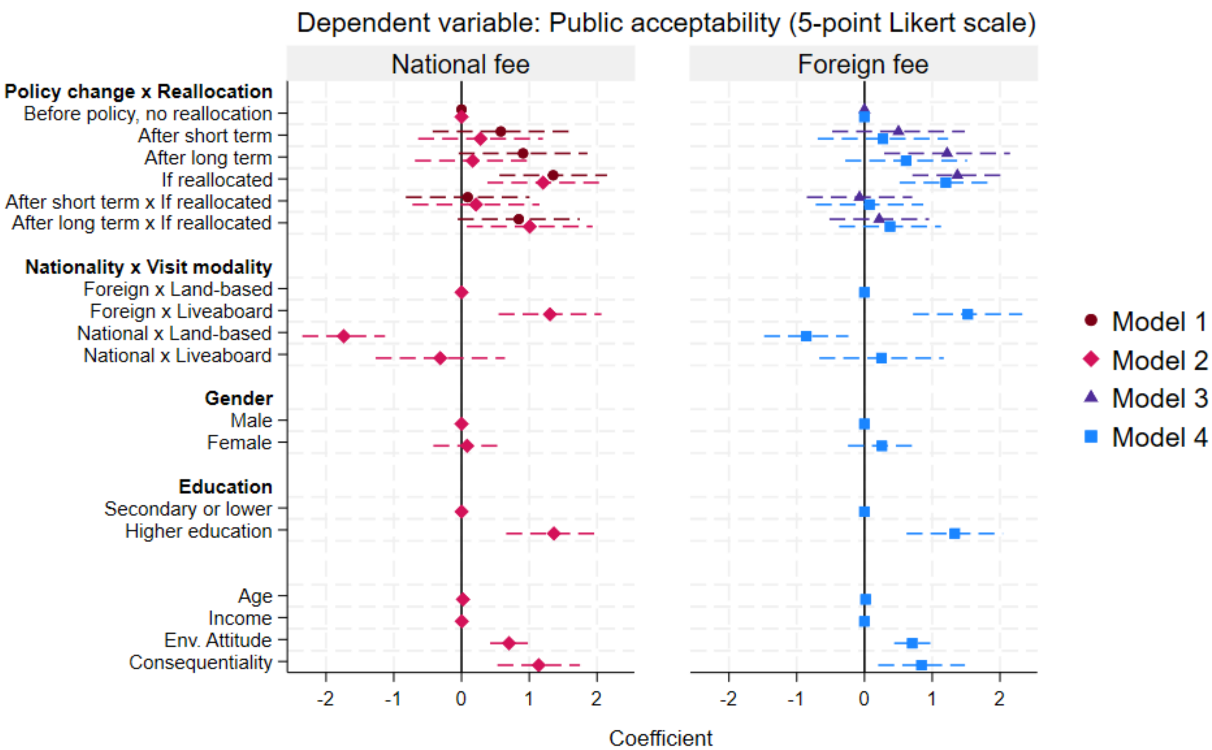

Figure 10: Tourists results - Estimates of public acceptability long after policy change.
Notes: This figure presents estimated coefficients and 95% confidence intervals (dashed lines) from mixed-effects ordered logit models of tourists' support for increasing entrance fees to the Galápagos National Park, before, shortly after (one-month) and long after (nine-month) policy implementation. Models 1–2 refer to support for increasing fees for domestic tourists, while Models 3–4 refer to fees for foreign tourists. Models 2 and 4 controls for demographics, visit modality, environmental attitudes, and perceived survey consequentiality. Regression tables are reported in the Appendix, Table A.11

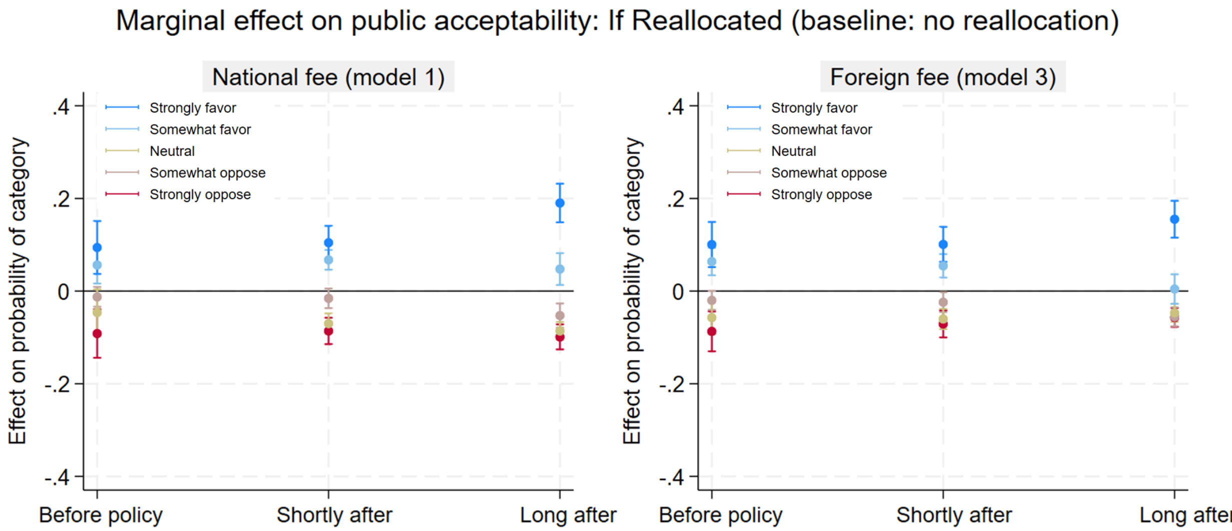

Figure 11: Tourists results - Marginal effects of revenues reallocation on public acceptability by policy change period.

Notes: This figure reports the marginal effect of the revenue reallocation on tourists' acceptability of increasing entrance fees to the Galápagos National Park, compared to the baseline scenario (no reallocation) across three periods: before, shortly after (one-month) and long after (nine-months) policy implementation. Estimates are based on Models 1 and 3, corresponding to increases in fees for domestic and foreign tourists, respectively. Regression tables are reported in the Appendix, Table A.11.

Overall, opposition to higher entrance fees is persistent when revenues are not explicitly linked to tangible benefits and institutional trust is weak, especially among residents. However, reallocating revenues toward publicly preferred uses substantially increases policy acceptability. These patterns are consistent with the dynamics described in the theoretical framework and with narratives indicating that residents evaluate entrance fee policies through expectations about transparency, fairness, and whether conservation revenues address local community needs.

6 Discussion and conclusion

6.1 General discussion

Our study examined how aligning revenue allocation with stakeholder preferences can improve support for a highly contested policy, an increase in conservation entrance fees, and how these attitudes evolve following policy implementation. We show that, despite strong initial opposition among residents, directing revenues toward publicly preferred uses substantially increases policy acceptability. The probability of strong support rises by around 10% before the policy and reaches up to 50% nine months after implementation. These findings offer a practical pathway for improving the political feasibility of contested policies for managing and financing protected areas. Given that delayed or ineffective implementation can accelerate environmental degradation, strengthening policy support is critical for reducing enforcement frictions and enabling timely adjustments of nature-based tourism policies, ultimately aimed at conserving biodiversity and sustaining local livelihoods.

In the Galápagos Islands, persistent opposition among residents reflects concerns about economic impacts, institutional distrust, and perceived inequalities in tourism governance. Follow-up survey responses indicate that many residents perceived a decline in visitation consistent with their pre-policy expectations. However, observed visitation trends were heterogeneous. During the first half of 2025, total arrivals increased by 2%, driven by a 15% rise in foreign visitors and a 15% decline in domestic visitors (Observatorio de Turismo de Galápagos, 2025). Because domestic tourists are predominantly land-based and more visible within local economies, this shift in demand composition likely shaped perceptions of overall decline despite stable aggregate arrivals.

The stability of foreign demand is consistent with survey findings showing that foreign tourists were generally more supportive of higher fees and less sensitive to revenue allocation than domestic visitors. This pattern also aligns with residents narratives describing foreign demand as relatively inelastic. From a governance perspective, sustained acceptability among residents may depend less on total arrivals and more on whether institutions credibly and transparently translate increased revenues into locally valued benefits delivered with distributive fairness.

6.2 Theoretical implications

This study advances public acceptability research by identifying and empirically testing a mechanism through which revenue allocation increases support for a policy change. In contrast to the environmental taxation literature (Douenne & Fabre, 2020; Kallbekken & Sælen, 2011), we show that beliefs about policy effectiveness can affect public support in a different direction. In the context of carbon taxes, perceived effectiveness in reducing emissions is typically associated with higher acceptability, while skepticism about behavioral responses to the tax represents a key barrier to support (Baranzini & Carattini, 2017). We contribute to this literature by showing that perceived effectiveness can also reduce public acceptability when policy impacts are expected to impose local economic costs. In our setting, many residents believe that higher entrance fees will effectively reduce tourist visits, which they associate with income losses. As a result, effectiveness is interpreted as a negative economic impact, constituting a source of opposition. The relationship between perceived effectiveness and acceptability thus depends on how expected outcomes are distributed across stakeholders. In nature-based tourism settings, where residents bear indirect costs and perceive long-standing inequalities, institutional trust and an effective redistribution of revenues toward locally valued benefits become even more important at sustaining support for entrance fee policies.

A second contribution concerns the temporal dynamics of acceptability. Whereas much of the literature treats acceptability as a static attitude or focuses on implementation in the short-term (Carattini et al., 2018; Schuitema et al., 2010), our study shows that opposition can persist over long periods. In this case, persistent opposition reflects realized costs aligning with prior expectations.

Nevertheless, revenue reallocation becomes increasingly effective after policy implementation, particularly among residents, where it operates as a compensatory mechanism that offsets perceived losses and enhances public benefits. We also identify stakeholder heterogeneity driven by differences in exposure to costs and expected benefits. Foreign tourists remain largely unaffected by the fee increase, likely due to their lower probability of revisiting the destination. Domestic visitors exhibit intermediate sensitivity, as they are more likely to return and experience the outcomes associated with revenue use. Residents, in contrast, show the strongest sensitivity, as they are continuously affected by how revenues are allocated. Tourist responses are consistent with economic theory, suggesting that destinations with strong market power such as high-biodiversity areas, face relatively inelastic demand (Rosselló-Nadal & Sard, 2026).

Third, the study shows the importance of transparency and procedural legitimacy. While trust in public institutions is found positively associated with policy support (as in Rodriguez-Sanchez et al., 2018), our qualitative analysis indicate that low institutional credibility is linked to perceived limitations in implementation capacity, skepticism about policy motivations, and concerns about reinforcing inequalities. Together, these findings suggest that incorporating stakeholder preferences into revenue allocation enhances acceptability, but public support extends beyond cost-benefit evaluations, involving perceptions of legitimacy and transparency in resource management as prerequisites. This supports literature highlighting that bottom-up governance approaches may be more effective than top-down policy imposition, especially for policies governing tourism in protected areas (Perlaviciute & Squintani, 2020).

6.3 Practical implications

Our findings align with the argument that entrance fees are a central governance instrument in protected areas, rather than merely a financing mechanism (Mach et al., 2020). For policymakers, a key implication is that the political feasibility of entrance fee adjustments depends on transparent and credible strategies for regulating tourism and delivering tangible benefits to local communities.

In the Galápagos Islands, a transparent and fair redistribution of public revenues could help mitigate the political instability that has undermined conservation efforts for more than two decades (Burbano & Meredith, 2020; Castrejón & Defeo, 2023; King et al., 2025). Given residents' stated preferences, authorities should prioritize investments in public services, particularly safe water and solid waste management, alongside continued efforts in species conservation. Such investments may reduce tensions between residents and protected area managers about strategies to transition toward a zero-growth tourism model, as urged by the World Heritage Committee (World Heritage Centre, 2018). The prioritization of both public services and species conservation also suggests that residents recognize the importance of protecting fragile ecosystems on which their livelihoods depend, creating opportunities to build a shared vision across stakeholder groups. As the fee increase is expected to double revenues, efficient financial mechanisms will be critical to optimize expenditures and align them with stakeholder priorities (Consejo de Gobierno del Régimen Especial de Galápagos, 2024; Laruelle, 2021).

More broadly, strengthening governance could involve participatory budgeting processes for allocating entrance fee revenues and other funding sources. Co-creating revenue allocation strategies with local communities may be particularly important, as our research evidences that participatory and transparent approaches increase policy acceptability, which can facilitate effective and timely policy implementation. Such approaches are both normatively desirable and strategically important (King et al., 2025), as they can help restore institutional trust and facilitate implementation of other contested policies, including fishing gear restrictions and marine zoning (Castrejón & Defeo, 2023; Castrejón et al., 2024). Enhanced coordination among institutions sharing fee revenues is also essential for an effective long-term governance (Burbano et al., 2022). Finally, if a transition toward a zero-growth tourism model remains a policy objective, complementary tools beyond fee increases, such as differentiated pricing schemes, visitor quotas, or flight restrictions, should be considered. Evaluating the acceptability of these measures represents an important direction for future policy design.

6.4 Limitations and future research

Although within-subject designs are well suited for examining shifts in acceptability across policy scenarios, between-subject designs may be more appropriate for assessing entirely new policies where respondents lack familiarity with the baseline. Future research could integrate both designs to evaluate alternative revenue allocation strategies. While this study captured attitudes nine months after implementation, longer observation periods would enable assessment of how realized environmental and economic outcomes reshape acceptability beyond the first year.

Besides revenue allocation, beliefs about demand impacts were an important factor in residents' narratives. Future studies could experimentally test how information about tourism demand elasticity or observed visitation trends influences acceptability and whether beliefs mediate support, although prior evidence suggests limited mediation effects (García-Antúnez et al., 2023). Incorporating willingness-to-pay and willingness-to-accept approaches could further link acceptability to welfare measures. Finally, field experiments examining participatory budgeting or co-designed revenue allocation processes could identify institutional arrangements that are more effective for building and sustaining public support for conservation financing policies.

Ultimately, entrance fees represent only one component of broader challenges in the governance of protected areas and nature-based tourism (Mach et al., 2020). However, aligning revenue allocation with both conservation priorities and local needs offers a practical pathway for restoring legitimacy and building public support. By providing a framework for improving the acceptability of policy instruments in high-biodiversity destinations, this study contributes to debates on how to design politically feasible policies that support the governance and long-term success of protected areas.

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A Appendix

Table A.1: Balanced Sample characteristics - Local Residents

Variable		Before policy change	After policy one-month	After policy nine-month
Residence	Non-permanent	16%	15%	10%
Status	Permanent	84%	85%	90%
Gender	Male	53%	53%	53%
	Female	47%	47%	47%
Age	Mean	34.7	34.7	34.7
18 - 70 years	(std. dev.)	(12.6)	(10.6)	(10.8)
Education	Secondary or lower	64%	64%	64%
level	Higher education	36%	36%	36%
Monthly income	Mean	0.89k	0.89k	0.89k
USD 0.25k - 6k	(std. dev.)	(0.97)	(0.65)	(0.50)
Economic	Commerce	16%	15%	19%
sector	Tourism-related	47%	48%	43%
	Govt. or conservation	12%	12%	12%
	Other sectors	14%	14%	14%
	Students and none	12%	12%	12%
Policy change	Yes	96%	99%	89%
awareness	No	4%	1%	11%
Observations		215	215	215

Notes: The samples before policy change and shortly after (one-month) are weighted to match the characteristics of the sample long after policy change (nine-month); weights are calculated using entropy balance based on age, gender, income, education and economic sector.

Table A.2: Balanced Sample characteristics - Tourists

Variable		Before policy change	After policy one-month	After policy nine-months
Tourist nationality	Domestic	48%	48%	48%
	Foreigner	52%	52%	52%
Gender	Male	50%	50%	50%
	Female	50%	50%	50%
Age 18 - 83 years	Mean	41.1	41.1	41.1
	(std. dev.)	(15.6)	(14.9)	(13.7)
Education	Secondary or lower	12%	12%	12%
	Higher education	88%	88%	88%
Monthly income USD 0.3k - 150k	Mean	10.5k	10.5k	10.5k
	(std. dev.)	(26.9k)	(26.3k)	(27.4k)
Occupation	Private worker	55%	57%	62%
	Public employee	25%	22%	24%
	Retired	11%	9%	7%
	Other	9%	12%	7%
Tourist modality	Land-based	67%	67%	67%
	Liveaboard	33%	33%	33%
Lenght of stay 3 - 60 days	Mean	8.9	6.9	7.2
	(std. dev.)	(10.9)	(4.6)	(5.9)
Policy change awareness	Yes	46%	61%	38%
	No	54%	39%	62%
Observations		376	376	376

Notes: The samples before policy change and shortly after policy (one-month) are weighted to match the characteristics of the sample long after policy (nine-month); weights are calculated using entropy balance based on nationality, age, education, income and visit modality.

Table A.3: Residents' Public acceptability - Pearson's chi-squared test: Without revenues reallocation vs If revenues are reallocated

Public acceptability 5-point Likert scale	National fee increase: W/o funds reallocation	National fee increase: If funds were reallocated	Difference %	Pearson's chi-squared Ho: No difference
Strongly oppose	51%	16%	-35%	chi2 = 326.60
Somewhat oppose	16%	8%	-8%	p = 0.000
Neutral	17%	10%	-7%	
Somewhat favor	9%	27%	18%	
Strongly favor	7%	39%	32%	
Public acceptability 5-point Likert scale	**Foreign fee increase: W/o funds reallocation**	**Foreign fee increase: If funds were reallocated**	**Difference %**	**Pearson's chi-squared Ho: No difference**
Strongly oppose	49%	16%	-33%	chi2 = 261.94
Somewhat oppose	15%	8%	-7%	p = 0.000
Neutral	14%	10%	-4%	
Somewhat favor	12%	27%	-5%	
Strongly favor	10%	39%	29%	
Observations	629	629	-	-

Table A.4: Residents' Public acceptability - Pearson's chi-squared test: Before policy vs Shortly after vs Long after

Public acceptability 5-point Likert scale	National fee increase w/o funds reallocation Before policy	 Shortly after	Difference %	Pearson's chi-squared Ho: No difference
Strongly oppose	48%	48%	0%	chi2 = 4.992
Somewhat oppose	16%	24%	8%	p = 0.288
Neutral	17%	13%	-4%	
Somewhat favor	11%	9%	-3%	
Strongly favor	8%	7%	-1%	
Public acceptability 5-point Likert scale	**Foreign fee increase w/o funds reallocation Before policy**	** Shortly after**	**Difference %**	**Pearson's chi-squared Ho: No difference**
Strongly oppose	49%	40%	-9%	chi2 = 4.961
Somewhat oppose	17%	20%	3%	p = 0.291
Neutral	11%	14%	3%	
Somewhat favor	12%	16%	4%	
Strongly favor	11%	10%	-1%	
Public acceptability 5-point Likert scale	**Overall fee increase if funds were reallocated Before policy**	** Shortly after**	**Difference %**	**Pearson's chi-squared Ho: No difference**
Strongly oppose	29%	16%	-13%	chi2 = 27.215
Somewhat oppose	11%	7%	-4%	p = 0.000
Neutral	12%	14%	2%	
Somewhat favor	31%	25%	-6%	
Strongly favor	17%	38%	21%	
Observations	203	210	-	-
Public acceptability 5-point Likert scale	**National fee increase w/o funds reallocation Shortly after**	** Long after**	**Difference %**	**Pearson's chi-squared Ho: No difference**
Strongly oppose	48%	56%	8%	chi2 = 22.281
Somewhat oppose	24%	8%	-16%	p = 0.000
Neutral	13%	21%	8%	
Somewhat favor	8%	7%	-1%	
Strongly favor	7%	8%	1%	
Public acceptability 5-point Likert scale	**Foreign fee increase w/o funds reallocation Shortly after**	** Long after**	**Difference %**	**Pearson's chi-squared Ho: No difference**
Strongly oppose	40%	58%	18%	chi2 = 25.132
Somewhat oppose	20%	7%	-13%	p = 0.000
Neutral	14%	18%	4%	
Somewhat favor	16%	8%	-8%	
Strongly favor	10%	9%	-1%	
Public acceptability 5-point Likert scale	**Overall fee increase if funds were reallocated Shortly after**	** Long after**	**Difference %**	**Pearson's chi-squared Ho: No difference**
Strongly oppose	16%	5%	-11%	chi2 = 37.844
Somewhat oppose	7%	6%	-1%	p = 0.000
Neutral	14%	4%	-10%	
Somewhat favor	25%	24%	-1%	
Strongly favor	38%	61%	23%	
Observations	210	216	-	-

Table A.5: Tourists' Public acceptability - Pearson's chi-squared test: W/o funds reallocation vs If funds are reallocated

National Tourists				
Public acceptability 5-point Likert scale	National fee increase: W/o funds reallocation	National fee increase: If funds were reallocated	Difference %	Pearson's chi-squared Ho: No difference
Strongly oppose	24%	11%	-13%	chi2 = 76.150
Somewhat oppose	24%	10%	-14%	p = 0.000
Neutral	24%	23%	-1%	
Somewhat favor	17%	26%	9%	
Strongly favor	11%	30%	19%	
Observations	388	387	-	-
Foreign Tourists				
Public acceptability 5-point Likert scale	Foreign fee increase: W/o funds reallocation	Foreign fee increase: If funds were reallocated	Difference %	Pearson's chi-squared Ho: No difference
Strongly oppose	10%	3%	-7%	chi2 = 29.162
Somewhat oppose	12%	7%	-5%	p = 0.000
Neutral	20%	18%	-2%	
Somewhat favor	25%	24%	-1%	
Strongly favor	33%	48%	15%	
Observations	340	338	-	-

Table A.6: Domestic tourists' Public acceptability - Pearson's chi-squared test: Before policy vs Shortly after vs Long after

National tourists				
Public acceptability 5-point Likert scale	National fee increase w/o funds reallocation Before policy	 Shortly after	Difference %	Pearson's chi-squared Ho: No difference
Strongly oppose	43%	22%	-21%	chi2 = 11.220
Somewhat oppose	19%	19%	0%	p = 0.024
Neutral	12%	25%	13%	
Somewhat favor	14%	19%	5%	
Strongly favor	12%	15%	3%	
Public acceptability 5-point Likert scale	Overall fee increase if funds were reallocated Before policy	 Shortly after	Difference %	Pearson's chi-squared Ho: No difference
Strongly oppose	20%	9%	-11%	chi2 = 10.468
Somewhat oppose	18%	10%	-8%	p = 0.033
Neutral	16%	29%	13%	
Somewhat favor	24%	23%	-1%	
Strongly favor	22%	29%	7%	
Public acceptability 5-point Likert scale	National fee increase w/o funds reallocation Shortly after	 Long after	Difference %	Pearson's chi-squared Ho: No difference
Strongly oppose	22%	17%	-5%	chi2 = 7.622
Somewhat oppose	19%	30%	11%	p = 0.106
Neutral	24%	28%	4%	
Somewhat favor	19%	16%	-3%	
Strongly favor	15%	9%	-4%	
Public acceptability 5-point Likert scale	Overall fee increase if funds were reallocated Shortly after	 Long after	Difference %	Pearson's chi-squared Ho: No difference
Strongly oppose	9%	9%	-1%	chi2 = 2.217
Somewhat oppose	10%	8%	-3%	p = 0.696
Neutral	29%	23%	-7%	
Somewhat favor	23%	28%	5%	
Strongly favor	29%	32%	6%	
Observations	204	313	-	-

Table A.7: Foreign tourists' Public acceptability - Pearson's chi-squared test: Before policy vs Shortly after vs Long after

Foreign tourists				
Public acceptability 5-point Likert scale	Foreign fee increase w/o funds reallocation Before policy	 After one-month	Difference %	Pearson's chi-squared Ho: No difference
Strongly oppose	4%	17%	13%	chi2 = 6.485
Somewhat oppose	20%	15%	-5%	p = 0.166
Neutral	26%	19%	-7%	
Somewhat favor	20%	26%	6%	
Strongly favor	30%	23%	-7%	
Public acceptability 5-point Likert scale	Overall fee increase if funds were reallocated Before policy	 After one-month	Difference %	Pearson's chi-squared Ho: No difference
Strongly oppose	4%	5%	1%	chi2 = 3.431
Somewhat oppose	5%	10%	5%	p = 0.488
Neutral	25%	21%	-4%	
Somewhat favor	23%	32%	9%	
Strongly favor	43%	32%	-11%	
Public acceptability 5-point Likert scale	Foreign fee increase w/o funds reallocation After one-month	 After nine-months	Difference %	Pearson's chi-squared Ho: No difference
Strongly oppose	17%	7%	-10%	chi2 = 13.704
Somewhat oppose	15%	9%	-6%	p = 0.008
Neutral	19%	19%	0%	
Somewhat favor	26%	26%	0%	
Strongly favor	23%	38%	15%	
Public acceptability 5-point Likert scale	Overall fee increase if funds were reallocated After one-month	 After nine-months	Difference %	Pearson's chi-squared Ho: No difference
Strongly oppose	5%	1%	-4%	chi2 = 19.181
Somewhat oppose	10%	5%	-5%	p = 0.001
Neutral	21%	16%	-5%	
Somewhat favor	32%	21%	-11%	
Strongly favor	32%	57%	25%	
Observations	141	294	-	-

Table A.8: Mixed Effects Ordered Logit - Residents' Public Acceptability of increasing conservation entry fees

Dependent variable: Public acceptability 5-point Likert scale	(1) National fee increase	(2) National fee increase	(3) Foreign fee increase	(4) Foreign fee increase
Policy change × Reallocation				
Before policy, No reallocation	–	–	–	–
Shortly after	-0.170 (0.329)	-0.272 (0.309)	0.555 (0.314)*	0.491 (0.305)
If reallocated	1.770 (0.218)***	1.746 (0.219)***	1.583 (0.217)***	1.580 (0.221)***
Shortly after × If reallocated	1.707 (0.325)***	1.724 (0.331)***	0.854 (0.301)***	0.846 (0.307)***
Gender				
Male		–		–
Female		-0.190 (0.295)		-0.222 (0.279)
Education				
Secondary or lower		–		–
Higher education		-0.188 (0.294)		-0.093 (0.273)
Economic sector				
Other sectors		–		–
Tourism and commerce		-0.245 (0.342)		-0.289 (0.327)
Govt. and conservation		1.901 (0.515)***		1.644 (0.508)***
Students and none		0.880 (0.524)*		0.827 (0.475)*
Age		-0.242 (0.150)		-0.151 (0.140)
Income		0.678 (0.234)***		0.581 (0.229)**
Trust in public inst.		0.492 (0.155)***		0.484 (0.145)***
Env. Attitude		0.380 (0.231)*		0.426 (0.208)**
Consequentiality		0.110 (0.282)		-0.054 (0.266)
Var(μ_i)	6.262 (1.178)***	4.966 (0.982)***	5.041 (0.973)***	4.206 (0.832)***
Observations	822.	810	822	810
Log-likelidhood	-1136.821	-1089.518	-1168.457	-1122.144
AIC	2289.641	2215.036	2352.914	2280.287
BIC	2327.335	2299.583	2390.608	2364.834

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses. The sample before policy change has been weighted to match the characteristics of the sample shortly after (one-month); weights are calculated using entropy balance based on age, gender, income, education and economic sector.

Table A.9: Mixed Effects Ordered Logit - Tourists' Public Acceptability of increasing conservation entry fees

Dependent variable: Public acceptability 5-point Likert scale	(1) National fee increase	(2) National fee increase	(3) Foreign fee increase	(4) Foreign fee increase
Policy change × Reallocation				
Before policy, No reallocation	–	–	–	–
Shortly after	0.622 (0.499)	0.166 (0.463)	0.519 (0.478)	0.147 (0.467)
If reallocated	1.469 (0.362)***	1.320 (0.369)***	1.204 (0.321)***	1.033 (0.324)**
Shortly after × If reallocated	-0.011 (0.427)	0.134 (0.435)	-0.116 (0.378)	0.038 (0.382)
Nationality × Tourist modality				
Foreign × Land-based		–		–
Foreign × Liveaboard		1.187 (0.628)		1.189 (0.647)
Domestic × Land-based		-1.902 (0.473)***		-0.877 (0.476)
Domestic × Liveaboard		0.453 (0.824)		0.928 (0.765)
Gender				
Male		–		–
Female		0.220 (0.379)		0.349 (0.364)
Education				
Secondary or lower		–		–
Higher education		1.244 (0.522)*		1.444 (0.509)**
Age		0.012 (0.013)		0.015 (0.014)
Income		-0.009 (0.007)		-0.013 (0.006)*
Env. Attitude		0.453 (0.172)**		0.393 (0.151)**
Consequentiality		1.247 (0.494)*		1.147 (0.488)*
Var(μ_i)	11.267 (2.093)***	9.247 (1.771)***	9.966 (1.939)***	8.279 (1.655)***
Observations	669	652	672	654
Log-likelihood	-1141.714	-1066.767	-1165.216	-1100.696
AIC	2299.429	2167.533	2346.431	2235.392
BIC	2335.475	2243.694	2382.513	2311.605

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses. The sample before policy change has been weighted to match the characteristics of the sample shortly after (one-month); weights are calculated using entropy balance based on nationality, visit modality, gender, age and income.

Table A.10: Mixed Effects Ordered Logit - Long term Residents' Public Acceptability of increasing conservation entry fees

Dependent variable: Public acceptability 5-point Likert scale	(1) National fee increase	(2) National fee increase	(3) Foreign fee increase	(4) Foreign fee increase
Policy change × Reallocation				
Before policy, No reallocation	–	–	–	–
Shortly after	-0.175 (0.277)	-0.263 (0.266)	0.449 (0.272)*	0.417 (0.268)
Long after	-0.277 (0.277)	-0.778 (0.294)***	-0.250 (0.283)	-0.742 (0.299)**
If reallocated	1.361 (0.187)***	1.349 (0.190)***	1.290 (0.203)***	1.297 (0.207)***
Shortly after × If reallocated	1.546 (0.301)***	1.573 (0.307)***	0.788 (0.291)***	0.770 (0.297)***
Long after × If reallocated	3.053 (0.357)***	3.121 (0.364)***	2.879 (0.347)***	2.965 (0.356)***
Gender				
Male		–		–
Female		-0.046 (0.182)		-0.062 (0.178)
Education				
Secondary or lower		–		–
Higher education		-0.086 (0.189)		-0.043 (0.185)
Economic sector				
Other sectors		–		–
Tourism and commerce		0.054 (0.226)		-0.076 (0.221)
Govt. and conservation		1.908 (0.372)***		1.655 (0.371)***
Students and none		0.564 (0.332)*		0.768 (0.331)**
Age		-0.138 (0.093)		-0.023 (0.092)
Income		0.433 (0.179)**		0.441 (0.179)**
Trust in public inst.		0.309 (0.100)***		0.322 (0.100)***
Env. Attitude		0.453 (0.110)***		0.513 (0.098)***
Consequentiality		-0.029 (0.189)		-0.004 (0.183)
Past respondent		1.330 (0.531)**		1.049 (0.519)**
var(μ_i)	2.925 (0.577)***	2.261 (0.466)***	2.550 (0.525)***	2.074 (0.431)***
Observations	1252	1240	1252	1240
Log-likelihood	-1709.008	-1642.594	-1739.215	-1672.732
AIC	3438.016	3327.188	3498.429	3387.465
BIC	3489.341	3434.768	3549.754	3495.045

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses.The samples before policy change and shortly after (one-month) are weighted to match the characteristics of the sample long after policy change (nine-month); weights are calculated using entropy balance based on age, gender, income, education and economic sector.

Table A.11: Mixed Effects Ordered Logit - Long term Tourists' Public Acceptability of increasing conservation entry fees

Dependent variable: Public acceptability 5-point Likert scale	(1) National fee increase	(2) National fee increase	(3) Foreign fee increase	(4) Foreign fee increase
Policy change × Reallocation				
Before policy, No reallocation	–	–	–	–
Shortly after	0.581 (0.513)	0.280 (0.470)	0.502 (0.499)	0.273 (0.491)
Long after	0.909 (0.485)*	0.165 (0.434)	1.220 (0.474)**	0.616 (0.458)
If reallocated	1.353 (0.405)***	1.204 (0.420)***	1.372 (0.338)***	1.201 (0.348)***
Shortly after × If reallocated	0.091 (0.466)	0.213 (0.478)	-0.073 (0.397)	0.075 (0.405)
Long after × If reallocated	0.845 (0.460)*	1.006 (0.474)**	0.218 (0.375)	0.378 (0.384)
Nationality × Tourist modality				
Foreign × Land-based		–		–
Foreign × Liveaboard		1.306 (0.387)***		1.524 (0.412)***
Domestic × Land-based		-1.739 (0.312)***		-0.859 (0.318)***
Domestic × Liveaboard		-0.313 (0.487)		0.252 (0.468)
Gender				
Male		–		–
Female		0.082 (0.254)		0.257 (0.255)
Education				
Secondary or lower		–		–
Higher education		1.364 (0.359)***		1.331 (0.362)***
Age		0.022 (0.009)**		0.018 (0.009)*
Income		0.004 (0.006)		-0.000 (0.005)
Env. Attitude		0.701 (0.142)***		0.705 (0.136)***
Consequentiality		1.140 (0.311)***		0.843 (0.327)***
Var(μ_i)	11.780 (1.526)***	8.756 (1.157)***	11.058 (1.536)***	8.783 (1.231)***
N	1421	1404	1424	1406
Log-likelihood	-2717.825	-2554.045	-2736.814	-2603.826
AIC	5455.649	5146.091	5493.627	5245.651
BIC	5508.240	5245.785	5546.239	5345.373

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses. The samples before policy change and shortly after (one-month) are weighted to match the characteristics of the sample long after policy change (nine-month); weights are calculated using entropy balance based on nationality, visit modality, gender, age and income.